

Integrated Volatility Matrix Estimation with Nonparametric Eigenvalue Regularization

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Abstract

Many existing volatility matrix estimators are designed to eliminate the bias from market microstructure noise, but are only applied for a single or a few assets. When the number of assets p is large compared with the sample size n , trivially extending these univariate estimators is not advised since they are all modifications of a sample realized covariance matrix, which suffers from bias in its extreme eigenvalues under the framework $p/n \rightarrow c > 0$. Without any implicit assumptions on the structure of the true integrated volatility matrix, we propose a nonparametric eigenvalue regularization on the multi-scale (NER-MSRVM), the kernel (NER-KRVM) and the pre-averaging (NER-PRVM) realized volatility matrix estimators. We show that our regularization can shrink nonlinearly those extreme eigenvalues on all three estimators above under the framework $p/n \rightarrow c \geq 0$, and are positive definite in probability. The inclusion of $c = 0$ extends our results to the case $p = o(n)$. Incidentally, the bias-corrected versions of kernel and pre-averaging estimators, which have faster rate of convergence at $n^{-1/4}$ but are not guaranteed to be positive definite in Barndorff-Nielsen et al. (2011) and Christensen et al. (2010) respectively, are now regularized to be positive definite in probability, and we prove their rates of convergence to an “ideal” estimator under the spectral norm are also at $n^{-1/4}$ under $p/n \rightarrow c \geq 0$. Our results are extended to allow for pervasive factors with rates spelt out, and a jump-diffusion model for the log-price processes with jumps removed using the wavelet method proposed in Fan and Wang (2007). Simulation results support our theoretical insights and demonstrate the superior performance of the regularization. A set of real high-frequency data is also analyzed.

Key words and phrases. High frequency data; Microstructure noise; Non-synchronous trading; Volatility matrix estimation; Nonlinear shrinkage.

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1 Introduction

Estimation of integrated volatility matrix is an important problem in high frequency data analysis, especially in portfolio allocation and risk management when the number of assets considered is more than one. The problems of non-synchronous trading times and the contamination of market microstructure noise on tick-by-tick data have prompted many researchers to consider different ways to efficiently estimate the integrated volatility matrix with the bias from such contamination removed. See for instance Aït-Sahalia et al. (2010) and Xiu (2010) for using maximum likelihood approaches, Griffin and Oomen (2011) for studying various existing estimators at the time assuming independent and identically distributed microstructure noise, while Zhang (2011) uses a multi-scale estimator to remove the bias contributed from it, and uses an all-refresh time scheme with previous-tick times to overcome the problem of non-synchronous trading times. All-refresh times are defined as the time points where all assets are traded at least once starting from a previous all-refresh time point, while a previous-tick time for an asset is the last trading time before an all-refresh time. For other refresh-time schemes, see for example Fan et al. (2012).

Focusing on the all-refresh times and using previous-tick times, a multi-scale realized volatility matrix (MSRVM) (Zhang, 2011), a kernel realized volatility matrix (KRVM) (Barndorff-Nielsen et al., 2011) and a pre-averaging realized volatility matrix (PRVM) (Christensen et al., 2010) are proposed and analyzed in the respective papers in order to remove the bias of microstructure noise on a realized volatility matrix. These three different estimators are all consistent at different element-wise rates of convergence, but the number of assets p is assumed fixed in all cases. When p is growing with the sample size n , in particular when $p/n \rightarrow c > 0$, it is inevitable that their performances will be poorer since all three estimators are still basically a realized volatility matrix, albeit modified to remove the bias from microstructure noise. As pointed out in Lam and Feng (2018), even in the simplest case of independent and identically distributed random vectors, a large sample covariance matrix will suffer from severe bias in its extreme eigenvalues under the framework $p/n \rightarrow c > 0$ (interested readers can see more details in their paper).

Kim et al. (2016) also point out that all three estimators are inconsistent when $p/n \rightarrow c > 0$. Under the assumption of sparseness of the true integrated volatility matrix, they propose to threshold the estimators and prove that a proper thresholding scheme results in consistent estimators for all three methods. At the same time, Dai et al. (2017) improves on a pre-averaging volatility matrix estimator by assuming an underlying factor structure in the log-price processes, resulting in a low-rank plus sparse estimator, and proved consistency with rates of convergence spelt out under a number of scenarios.

In this paper, we propose nonparametric eigenvalue regularized MSRVM, KRVM and PRVM, abbreviated as NER-MSRVM, NER-KRVM and NER-PRVM respectively, following a successful application on a

two-scaled volatility matrix estimator in Lam and Feng (2018). The multi-scaled estimator NER-MSRVM in particular is a generalization of the two-scaled estimator proposed in Lam and Feng (2018). Our contributions are three-fold. First of all, our estimators are flexible to use since they do not need any structural assumptions on the true integrated volatility matrix or the log-price processes as opposed to those in Kim et al. (2016) and Dai et al. (2017). See Section 6 also where we have demonstrated empirically that our estimators outperform other state-of-the-art competitors.

Secondly, although without any structural assumptions we are not able to prove consistency, we proved that all estimators are positive-definite in probability with their extreme eigenvalues regularized, and provide rate of convergence in spectral norm of each estimator to a rotation-equivariant “ideal” estimator, to be defined in Section 3.4, all under the framework $p/n \rightarrow c \geq 0$. The inclusion of $c = 0$ extends our results to the case when $p = o(n)$, which is also interesting when there are pervasive factors in the log-price processes. For NER-MSRVM, when there are no pervasive factors, it turns out that without sparsity assumption, the spectral norm rate of convergence is independent of p and is at $n^{-1/6}$ only when we modify the largest and the number of scales of the estimator from $n^{1/2}$ used in Kim et al. (2016), to $n^{2/3}$. With NER-KRVM and NER-PRVM, when using the positive definite versions as in Barndorff-Nielsen et al. (2011) and Christensen et al. (2010) respectively, their rates of convergence to the ideal estimator are both $n^{-1/5}$, while it is $n^{-1/4}$ when using their bias-corrected versions. These rates of convergence are the same as in the respective papers, but we are working in the framework $p/n \rightarrow c > 0$. More importantly, while the bias-corrected versions of KRVM and PRVM which converge at a rate of $n^{-1/4}$ are not guaranteed to be positive semi-definite, we prove in Theorem 2 and 3 that their corresponding regularized versions, NER-KRVM and NER-PRVM, are both positive definite in probability. Our results also allow pervasive factors to be included in the log-price processes, and all estimators will then have slower rate of convergence. In general we then need $p = o(n)$ (i.e., $c = 0$) for guaranteed convergence of our estimators to their ideal counterpart. See Section 4 for more details.

Finally, we show that with wavelet jumps-removal proposed in Fan and Wang (2007), all rates of convergence remain the same. We have also presented a theorem on minimum variance portfolio using our regularized estimators in Section 6.2.1, showing that the constructed portfolios using our estimators satisfy a maximum exposure bound, which is shared by the theoretical minimum variance portfolio.

The rest of the paper is organized as follows. Section 2 presents all necessary notations and the model used for a log-price process. Challenges for analyzing high-frequency data are also explained. Section 3 introduces MSRVM, KRVM and PRVM. How nonlinear shrinkage works on these three estimators are detailed in Section 3.4. Asymptotic theories and detailed assumptions, including those with jumps-removal in the case of jump-diffusion log-price processes, can be found in Section 4. Practical concerns and imple-

mentation can be found in Section 5, while all simulations and a thorough empirical study are presented in Section 6.1. All proofs of theorems are presented in the supplementary materials for the paper.

2 Model and Notations

2.1 Price model

Let $\mathbf{X}_t = (X_t^{(1)}, X_t^{(2)}, \dots, X_t^{(p)})^T$ be a vector of log prices of p assets at time t , which follows a continuous-time diffusion model

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \boldsymbol{\sigma}_t d\mathbf{W}_t, \quad t \in [0, 1], \quad (2.1)$$

with $\{\mathbf{W}_t\}$ being a p -dimensional standard Brownian motion and $\boldsymbol{\mu}_t$ a càdlàg (element-wise) process, which can be random and correlated with $\{\mathbf{W}_t\}$ in general. The volatility $\boldsymbol{\sigma}_t \in \mathbb{R}^{p \times p}$ is also càdlàg. For a period $[a, b]$ where $0 \leq a \leq b \leq 1$, the corresponding integrated volatility matrix is defined as

$$\boldsymbol{\Sigma}(a, b) = \int_a^b \boldsymbol{\sigma}_u \boldsymbol{\sigma}_u^T du.$$

The integrated volatility matrix is an important input in risk management. For instance, if we have a portfolio $\mathbf{w}$ which stays constant over the period $[a, b]$, then the accumulated risk of the portfolio over the period is

$$R^{1/2}(\mathbf{w}) = (\mathbf{w}^T \boldsymbol{\Sigma}(a, b) \mathbf{w})^{1/2} = \left(\int_a^b \mathbf{w}^T \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^T \mathbf{w} dt \right)^{1/2},$$

where the term $\mathbf{w}^T \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^T \mathbf{w}$ can be considered an instantaneous square-risk at time t .

2.2 Data Splitting

Similar to Lam and Feng (2018), our regularized estimators are defined over a partition of the normalized time period $[0, 1]$. Let L be the total number of partitions, and

$$0 = \tau_0 < \tau_1 < \dots < \tau_L = 1,$$

where $(\tau_{\ell-1}, \tau_\ell]$ denotes the ℓ th partition. See Section 3 for the definition of all estimators, including our regularized estimators which utilize the partition above.

2.3 Asynchronicity and microstructure noise

To deal with the problem of asynchronous trading times, we use the method of all-refresh times. See Zhang (2011) for instance. An all-refresh time is the time until all assets are traded at least once from a past all-refresh time point. Let $n(\ell)$ be the number of all-refresh time points in the partition $(\tau_{\ell-1}, \tau_\ell]$, $\ell = 1, \dots, L$, so that the total number of all-refresh time points is nL , where $n = L^{-1} \sum_{\ell=1}^L n(\ell)$ is the average number of all-refresh time points in each partition. In this paper, we assume n is the same order as $n(\ell)$ for all ℓ , and is also the same order as nL since L is assumed finite throughout the paper.

Let $\{v_s\}_{s=1, \dots, nL}$ be the set of all-refresh times in $[0, 1]$ for $\mathbf{X}_t$. Then all estimators in this paper are calculated based on the previous-tick times $t_s^j \in (v_{s-1}, v_s]$, $s = 1, \dots, nL$, $j = 1, \dots, p$, which is the last trading time for the j th asset before time v_s . Note that $t_s^{j_1} \neq t_s^{j_2}$ for $j_1 \neq j_2$ in general.

With these notations, we write the observed prices that we are using for the construction of our estimators as

$$\mathbf{Y}(s) = \mathbf{X}(s) + \boldsymbol{\epsilon}(s), \quad s = 1, \dots, nL, \quad (2.2)$$

where $\{\boldsymbol{\epsilon}_t\}$ is the process of microstructure noise, with $\mathbf{X}(s) = (X_{t_s^1}^{(1)}, \dots, X_{t_s^p}^{(p)})^\top$ and $\boldsymbol{\epsilon}(s) = (\epsilon_{t_s^1}^{(1)}, \dots, \epsilon_{t_s^p}^{(p)})^\top$. Note that $\boldsymbol{\epsilon}(\cdot)$ can be dependent on $\mathbf{X}(\cdot)$ in general.

3 Integrated Volatility Matrix Estimators

We first introduce three existing integrated volatility matrix estimators before presenting their regularized versions. They are the multi-scale realized volatility matrix estimator (MSRVM) (extended from the univariate version of Zhang (2006)), the kernel realized volatility matrix estimator (KRVM) introduced in Barndorff-Nielsen et al. (2011) and the pre-averaging realized volatility matrix estimator (PRVM) by Christensen et al. (2010). They all have an element-wise rate of convergence of $n^{-1/4}$, which is faster than the rate $n^{-1/6}$ for the two-scale realized volatility matrix estimator (TSRVM) analyzed in Zhang (2011). Kim et al. (2016) shows that the positive-definite versions of KRVM and PRVM can still achieve an element-wise rate $n^{-1/5}$, which is still faster than $n^{-1/6}$.

Lam and Feng (2018) introduces the nonparametric eigenvalue regularized integrated volatility matrix estimator (NERIVE) which is based on a modification of the TSRVM. As their spectral norm rate of convergence to an “ideal” estimator (see Section 3.4 for more details) is only $n^{-1/6}$, the same as the element-wise rate for TSRVM, we analyze in Section 4 if our eigenvalue regularization can be applied on MSRVM, KRVM and PRVM to achieve better rates of convergence and improved practical performance.

3.1 Multi-scale realized volatility matrix

Zhang (2006) proposes a multi-scale approach in stochastic volatility estimation to eliminate the bias from microstructure noise. Our method is based on the multi-scale realized volatility matrix (MSRVM) analyzed in Tao et al. (2013). For $\ell = 1, \dots, L$, define the MSRVM on each partition $(\tau_{\ell-1}, \tau_\ell]$ to be

$$\begin{aligned} \mathbf{MS}(\mathbf{Y})_\ell &= \sum_{m=1}^M a_m [\mathbf{Y}, \mathbf{Y}^\top]_\ell^{(K_m)} + \zeta \left([\mathbf{Y}, \mathbf{Y}^\top]_\ell^{(K_1)} - [\mathbf{Y}, \mathbf{Y}^\top]_\ell^{(K_M)} \right), \text{ where} \\ [\mathbf{Y}, \mathbf{Y}^\top]_\ell^{(m)} &= \frac{1}{m} \sum_{s \in S^\ell(m)} (\mathbf{Y}(s) - \mathbf{Y}(s-m))(\mathbf{Y}(s) - \mathbf{Y}(s-m))^\top, \\ S^\ell(m) &= \{s : t_s^i, t_{s-m}^i \in (\tau_{\ell-1}, \tau_\ell] \text{ for all } i\}, \quad |S^\ell(m)|_m = \frac{|S^\ell(m)| - m + 1}{m}, \\ K_m &= N + m, \quad a_m = \frac{12(m+N)(m-M/2-1/2)}{M(M^2-1)}, \quad \zeta = \frac{(M+N)(N+1)}{(n+1)(M-1)}, \end{aligned} \quad (3.3)$$

with $N \asymp n^{2/3}$ and $M \asymp n^{1/2}$, where $a \asymp b$ means that $a = O(b)$ and $b = O(a)$. For the ease of presentation of our regularized estimators in Section 3.4, we also use the notation

$$\tilde{\Sigma}(\tau_{\ell-1}, \tau_\ell)^M = \mathbf{MS}(\mathbf{Y})_\ell. \quad (3.4)$$

The above estimator is different from the one considered in Tao et al. (2013) in that we have set $N \asymp n^{2/3}$ rather than $n^{1/2}$ which is used in Tao et al. (2013) and Kim et al. (2016). The parameter N controls the magnitude of the scales used in the estimator. The reason that we use a different magnitude of scale is that unlike Tao et al. (2013) and Kim et al. (2016), we do not assume sparsity of the underlying integrated volatility matrix. Without this assumption, a diverging p can significantly increase the bias contributed from microstructure noise and asynchronous transactions. In the end, our proofs reveal that only a scale of magnitude $N \asymp n^{2/3}$ can remove the bias effects from a diverging p of the same order as n , even with our eigenvalue regularization to be presented in Section 3.4. The final rate of convergence then goes from $n^{-1/4}$, the best univariate rate, to $n^{-1/6}$. See Theorem 1 in Section 4 for more details. Incidentally, this scale magnitude is what is used in Lam and Feng (2018), leading to the same rate of convergence $n^{-1/6}$ for the nonparametric eigenvalue regularized two-scale realized volatility matrix estimator.

In Section 6, the performance of the corresponding regularized estimator using $N \asymp n^{2/3}$ is also compared to that using $N \asymp n^{1/2}$, and it is clear that the one using $N \asymp n^{2/3}$ has a better practical performance.

Remark 1. Two important conditions to be satisfied for the MSRVM are that $\sum_{m=1}^M (a_m/K_m) = 0$ and $\sum_{m=1}^M a_m = 1$ (Tao et al., 2013, Zhang, 2006). These conditions ensure the estimator has bias contributed

from the microstructure noise removed, and is unbiased, respectively, resulting at an element-wise rate of convergence $n^{-1/4}$. Zhang (2006) also shows that efficiency can be improved by combining more than two time scales. This is also the reason why our regularized estimator based on MSRV should perform better than the corresponding two-scale version, which is supported by our simulations in Section 6. However, there is no choice of a_m such that a positive semi-definite estimator is guaranteed. With our nonparametric eigenvalue regularization though, we can prove positive definiteness in probability for our regularized estimator under $p/n \rightarrow c \geq 0$. See Theorem 1 in Section 4 for more details.

3.2 Kernel realized volatility matrix

Barndorff-Nielsen et al. (2011) proposes a multivariate realized kernel estimator which does not smooth the covariance but the autocovariance operators. The estimator is robust to microstructure noise and can handle asynchronous trading. It also has a positive semi-definite version with a slightly different definition. They introduce jittering, which is to average J observations at the beginning and the end of a period. With J going to infinity at an appropriate rate, the error at the beginning and the end of the day is averaged away. We adopt this scheme and denote the jittered log-price vector as $\mathbf{Y}_\ell^{(J)}(s)$, $s = 1, \dots, n(\ell) - 2J + 1$, $\ell = 1, \dots, L$. Define

$$\begin{aligned} \mathbf{K}(\mathbf{Y})_\ell &= \gamma_\ell^{(0)}(\mathbf{Y}_\ell^{(J)}) + \sum_{h=1}^{n(\ell)-2J} k\left(\frac{h-1}{H}\right) [\gamma_\ell^{(h)}(\mathbf{Y}_\ell^{(J)}) + \gamma_\ell^{(-h)}(\mathbf{Y}_\ell^{(J)})], \quad \text{where} \\ \gamma_\ell^{(h)}(\mathbf{Y}_\ell^{(J)}) &= \sum_{s=h+2}^{n(\ell)-2J+1} (\mathbf{Y}_\ell^{(J)}(s) - \mathbf{Y}_\ell^{(J)}(s-1))(\mathbf{Y}_\ell^{(J)}(s-h) - \mathbf{Y}_\ell^{(J)}(s-h-1))^\top, \quad h \geq 0, \end{aligned} \quad (3.5)$$

with $k(\cdot)$ a kernel function and H a bandwidth parameter. We also define $\gamma_\ell^{(h)}(\mathbf{Y}_\ell^{(J)}) = \gamma_\ell^{(-h)}(\mathbf{Y}_\ell^{(J)})^\top$ for $h < 0$, and assume that (i) $k(0) = 1$ and $k'(0) = 0$; (ii) $k(\cdot)$ is twice differentiable with continuous derivatives; (iii) $\int_0^\infty k(x)^2 dx$, $\int_0^\infty k'(x)^2 dx$ and $\int_0^\infty k''(x)^2 dx$ are finite. We also use the notation

$$\tilde{\Sigma}(\tau_{\ell-1}, \tau_\ell)^K = \mathbf{K}(\mathbf{Y})_\ell. \quad (3.6)$$

For the positive semi-definite version, assume further (iv) $\int_{-\infty}^\infty k(x) \exp(ix\lambda) dx \geq 0$ for all $\lambda \in \mathbb{R}$. Then the positive semi-definite KRVM for the partition $(\tau_{\ell-1}, \tau_\ell]$ is defined as

$$\tilde{\Sigma}(\tau_{\ell-1}, \tau_\ell)^{KP} = \gamma_\ell^{(0)}(\mathbf{Y}_\ell^{(J)}) + \sum_{h=1}^{n(\ell)-2J} k\left(\frac{h}{H}\right) [\gamma_\ell^{(h)}(\mathbf{Y}_\ell^{(J)}) + \gamma_\ell^{(-h)}(\mathbf{Y}_\ell^{(J)})]. \quad (3.7)$$

3.3 Pre-averaging realized volatility matrix

The idea of pre-averaging is to smooth the high-frequency data first. The KRVM in (3.6) is also a kind of smoothing mechanism, but it only smooths the autocovariance operators instead of the data. The data smoothing idea is first presented in Hayashi et al. (2005), and Christensen et al. (2010) extends it to the case with microstructure noise by using pre-averaging and show that the resulting estimator is consistent. This estimator has the property that it can be implemented directly on irregular, asynchronous and noisy observations without any form of imputation.

We adopt a different bias-correction term than Christensen et al. (2010) used, which is needed for our regularization to be introduced in Section 3.4 to work. Let $\mathbf{Y}_\ell(j)$ be the j th all-refresh log-price vector within the ℓ th partition, $j = 1, \dots, n(\ell)$. Then define

$$\begin{aligned} \mathbf{P}(\mathbf{Y})_\ell &= \frac{1}{\psi Q} \sum_{j=1}^{n(\ell)-Q+1} [\bar{\mathbf{Y}}_j^{(\ell)} \bar{\mathbf{Y}}_j^{(\ell)\top} - \varsigma \hat{\boldsymbol{\eta}}^{(\ell)}], \quad \text{where} \\ \hat{\boldsymbol{\eta}}^{(\ell)} &= \frac{1}{2n(\ell)} \sum_{s=2}^{n(\ell)} (\mathbf{Y}_\ell(s) - \mathbf{Y}_\ell(s-1))(\mathbf{Y}_\ell(s) - \mathbf{Y}_\ell(s-1))^\top, \\ \bar{\mathbf{Y}}_j^{(\ell)} &= \sum_{l=1}^{Q-1} g\left(\frac{l}{Q}\right) [\mathbf{Y}_\ell(j+l) - \mathbf{Y}_\ell(j+l-1)], \\ \varsigma &= \sum_{l=0}^{Q-1} \left[g\left(\frac{l}{Q}\right) - g\left(\frac{l+1}{Q}\right) \right]^2, \quad \psi = \int_0^1 g(t)^2 dt, \end{aligned} \tag{3.8}$$

with Q a bandwidth parameter of order $n^{1/2}$. The function $g(\cdot)$ is continuous and piecewise continuously differentiable with a piecewise Lipschitz derivative g' , satisfying $g(0) = g(1) = 0$ and $\int_0^1 g(t)^2 dt > 0$. We also use the notation

$$\tilde{\boldsymbol{\Sigma}}(\tau_{\ell-1}, \tau_\ell)^P = \mathbf{P}(\mathbf{Y})_\ell. \tag{3.9}$$

The diagonal elements in $\hat{\boldsymbol{\eta}}^{(\ell)}$ can certainly be replaced by those defined in Christensen et al. (2010) which will then be more accurate since it used up all available tick-by-tick data, but we do need off-diagonal elements to be non-zero as well, and $\hat{\boldsymbol{\eta}}^{(\ell)}$ defined above is sufficient to give us the best rate for our regularized estimator.

For the positive semi-definite version, with Q of order $n^{3/5}$, define

$$\tilde{\boldsymbol{\Sigma}}(\tau_{\ell-1}, \tau_\ell)^{PP} = \frac{1}{\psi Q} \sum_{j=1}^{n(\ell)-Q+1} \bar{\mathbf{Y}}_j^{(\ell)} \bar{\mathbf{Y}}_j^{(\ell)\top}. \tag{3.10}$$

3.4 Nonparametric eigenvalue regularization

Multi-scale, kernel and pre-averaging realized volatility matrix estimators perform well in terms of removing the bias contributed from microstructure noise. Yet under the setting $p/n \rightarrow c > 0$, a typical realized volatility matrix suffers from bias in its extreme eigenvalues. Also, the spread of the eigenvalues of the estimator is typically much larger than its population counterpart. This creates instability in applications. Since we do not have any structural assumptions like sparsity of the underlying matrix, regularization with sparsity such as those introduced in Kim et al. (2016) cannot be used.

This paper follows a branch of regularization methods which focus on shrinking the extreme eigenvalues of a large covariance matrix estimator. The simplest one is linear shrinkage introduced in Ledoit and Wolf (2004), which shrinks a large sample covariance matrix towards a target matrix. Ledoit and P  ch   (2011) proves a nonlinear shrinkage formula for shrinking the eigenvalues of a large sample covariance matrix for independent and identically distributed data. This leads to an estimator based on this nonlinear shrinkage method in Ledoit and Wolf (2012). On the other hand, Abadir et al. (2014) proposes to split the data into two parts under a setting with stationary covariance matrix for identically distributed data. Under the setting $p/n \rightarrow c > 0$, Lam (2016) shows that if the data is split under a certain ratio, then the extreme eigenvalues of the sample covariance matrix are shrunk nonlinearly asymptotically, following the same asymptotic formula for the shrunk eigenvalues proved in Ledoit and P  ch   (2011). Lam et al. (2017) adapts such a data splitting regularization for intra-day data (not tick-by-tick) without contamination of microstructure noise on the log-price, and proposed an estimator with shrunk extreme eigenvalues.

The above methods all assume some form of independence for the data in proving asymptotic results. For high-frequency data with contamination from microstructure noise, we cannot assume independence of the log-returns. A breakthrough comes from Lam and Feng (2018) which utilizes a data splitting scheme the same as in Section 2.2 to perform eigenvalue regularization for the TSRVM. They do not assume serial independence of microstructure noise, which can also be cross-sectionally correlated in general. All log-prices can also be correlated with the microstructure noise too. Their estimator is positive definite in probability under the setting $p/n \rightarrow c > 0$, and the spread of eigenvalues is shrunk within that for the population integrated volatility matrix. With these advantages, below we introduce a similar regularization scheme for the MSRVM, KRVM and PRVM. As discussed in the Introduction, we hope to achieve a rate of convergence faster than $n^{-1/6}$, the rate proved in Lam and Feng (2018). We also allow for $c = 0$ in our theorems so that $p = o(n)$ is allowed, which is in general needed for our estimators to be converging to an ideal estimator when pervasive factors exist in the log-price processes.

First, using the data splitting scheme in Section 2.2, we introduce a rotation-equivariant estimator

$\Sigma(\mathbf{D}) = \mathbf{P}_{-j} \mathbf{D} \mathbf{P}_{-j}^T$, where $\mathbf{D}$ is a diagonal matrix, and $\mathbf{P}_{-j}$ is orthogonal such that

$$\tilde{\Sigma}_{-j} = \mathbf{P}_{-j} \mathbf{D}_{-j} \mathbf{P}_{-j}^T, \quad j = 1, \dots, L, \quad \text{with } \tilde{\Sigma}_{-j} = \sum_{\ell \neq j} \tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell}). \quad (3.11)$$

Here, $\tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})$ can be $\tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})^M$, $\tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})^K$ or $\tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})^P$ in (3.4), (3.6) and (3.9) respectively, or $\tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})^{KP}$, $\tilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})^{PP}$ in (3.7) and (3.10) for the positive semi-definite versions of KRVM and PRVM respectively. This class of estimators is used in Ledoit and Wolf (2012), Lam (2016) and Lam and Feng (2018) as a starting point for the construction of their regularized estimators. First introduced in James and Stein (1961) for estimating a covariance matrix under the Stein's loss function, this class is invariant under rotation, and serves as a good starting point with no *a priori* information of the eigenvectors of the population covariance matrix.

Next, we want to bring $\tilde{\Sigma}_{-j}$ to be as close to $\Sigma(\tau_{j-1}, \tau_j)$ as possible by fixing its eigenvectors but varying its eigenvalues. This essentially makes it a p -dimensional problem. To do this, we consider the following optimization problem:

$$\min_{\mathbf{D} \text{ diagonal}} \|\mathbf{P}_{-j} \mathbf{D} \mathbf{P}_{-j}^T - \Sigma(\tau_{j-1}, \tau_j)\|_F, \quad (3.12)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The same problem is also considered in Lam and Feng (2018). The reason to use $\mathbf{P}_{-j}$ for the rotation-equivariant class is that we can condition all information outside partition j ($\mathbf{P}_{-j}$ is then fixed), so that then the correlation between $\{\mathbf{X}_t\}$ and $\{\epsilon_t\}$, and the serial correlation in $\{\epsilon_t\}$ within partition j can be weakened. See Assumption (E3) in Section 4. If each partition is “small”, then each $\mathbf{P}_{-j}$ is “close” to each other intuitively since each $\mathbf{P}_{-j}$ uses up all information except those within the partition.

The solution for (3.12) is $\mathbf{D} = \text{diag}(\mathbf{P}_{-j}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{P}_{-j})$, where $\text{diag}(A)$ is a diagonal matrix with the diagonal elements of A . See Proposition 1 in Lam and Feng (2018) and the proof therein. It is not difficult to see that the spread of eigenvalues of such $\mathbf{D}$ is constrained within that of $\Sigma(\tau_{j-1}, \tau_j)$. In Section 4, we present in Theorem 1, 2 and 3 that in spectral norm, $\text{diag}(\mathbf{P}_{-j}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^M \mathbf{P}_{-j})$, $\text{diag}(\mathbf{P}_{-j}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^K \mathbf{P}_{-j})$, $\text{diag}(\mathbf{P}_{-j}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^P \mathbf{P}_{-j})$, $\text{diag}(\mathbf{P}_{-j}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^{KP} \mathbf{P}_{-j})$ and $\text{diag}(\mathbf{P}_{-j}^T \tilde{\Sigma}(\tau_{j-1}, \tau_j)^{PP} \mathbf{P}_{-j})$ are converging to $\mathbf{D} = \text{diag}(\mathbf{P}_{-j}^T \Sigma(\tau_{j-1}, \tau_j) \mathbf{P}_{-j})$ in probability. Therefore, our regularized volatility matrix estimators of

MSRVM, KRVM and PRVM for the partition $(\tau_{j-1}, \tau_j]$ are respectively

$$\widehat{\Sigma}(\tau_{j-1}, \tau_j)^M = \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \widetilde{\Sigma}(\tau_{j-1}, \tau_j)^M \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top, \quad (3.13)$$

$$\widehat{\Sigma}(\tau_{j-1}, \tau_j)^K = \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \widetilde{\Sigma}(\tau_{j-1}, \tau_j)^K \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top, \quad (3.14)$$

$$\widehat{\Sigma}(\tau_{j-1}, \tau_j)^P = \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \widetilde{\Sigma}(\tau_{j-1}, \tau_j)^P \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top. \quad (3.15)$$

We also denote $\widehat{\Sigma}(\tau_{j-1}, \tau_j)^{KP}$ and $\widehat{\Sigma}(\tau_{j-1}, \tau_j)^{PP}$ the corresponding estimators using $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^{KP}$ and $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^{PP}$ respectively. The corresponding nonparametric eigenvalue regularized integrated volatility matrix estimators for the period $[0, 1]$ are then defined to be

$$\widehat{\Sigma}(0, 1)^M = \sum_{j=1}^L \widehat{\Sigma}(\tau_{j-1}, \tau_j)^M = \sum_{j=1}^L \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \widetilde{\Sigma}(\tau_{j-1}, \tau_j)^M \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top, \quad (3.16)$$

$$\widehat{\Sigma}(0, 1)^K = \sum_{j=1}^L \widehat{\Sigma}(\tau_{j-1}, \tau_j)^K = \sum_{j=1}^L \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \widetilde{\Sigma}(\tau_{j-1}, \tau_j)^K \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top, \quad (3.17)$$

$$\widehat{\Sigma}(0, 1)^P = \sum_{j=1}^L \widehat{\Sigma}(\tau_{j-1}, \tau_j)^P = \sum_{j=1}^L \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \widetilde{\Sigma}(\tau_{j-1}, \tau_j)^P \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top, \quad (3.18)$$

which are abbreviated as NER-MSRVM, NER-KRVM and NER-PRVM respectively. We also denote $\widehat{\Sigma}(0, 1)^{KP}$ and $\widehat{\Sigma}(0, 1)^{PP}$ the corresponding estimators using $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^{KP}$ and $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^{PP}$, and are abbreviated as NER-KRPVM and NER-PRPVM respectively. An ideal estimator relative to $\widehat{\Sigma}(0, 1)$ is an estimator with $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^M$, $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^K$ (or $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^{KP}$) and $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^P$ (or $\widetilde{\Sigma}(\tau_{j-1}, \tau_j)^{PP}$) replaced by the population counterpart $\Sigma(\tau_{j-1}, \tau_j)$,

$$\Sigma_{\text{Ideal}}(0, 1) = \sum_{j=1}^L \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \Sigma(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top. \quad (3.19)$$

As discussed in the paragraph after (3.12), if each partition is small, then say each $\mathbf{P}_{-j}$ is close to $\mathbf{P}$ which is the orthogonal matrix from the eigen-decomposition of $\sum_{\ell} \widetilde{\Sigma}(\tau_{\ell-1}, \tau_{\ell})$, then the above becomes

$$\Sigma_{\text{Ideal}}(0, 1) \approx \mathbf{P} \text{diag}(\mathbf{P}^\top \Sigma(0, 1) \mathbf{P}) \mathbf{P}^\top,$$

which resembles an ideal estimator for $\Sigma(0, 1)$ that utilizes all data information for constructing the eigenmatrix $\mathbf{P}$. Indeed such an ideal estimator is defined in Ledoit and Wolf (2012) and Lam (2016), and is treated as a benchmark for evaluating performances of different estimators. Note that if $\mathbf{P}$ is close to the eigenmatrix of $\Sigma(0, 1)$ (say, when $p/n \rightarrow 0$), then $\Sigma_{\text{Ideal}}(0, 1) \approx \Sigma(0, 1)$. As $p/n \rightarrow c > 0$ and $\mathbf{P}$ is getting further away from the true eigenmatrix of $\Sigma(0, 1)$, $\Sigma_{\text{Ideal}}(0, 1)$ in (3.19) is still close

to $\mathbf{P}\text{diag}(\mathbf{P}^\top \boldsymbol{\Sigma}(0,1)\mathbf{P})\mathbf{P}^\top$, the ideal rotation-equivariant estimator utilizing all information for the construction of $\mathbf{P}$.

In practice, the above is the reason why we want each partition to be as small as possible, but with enough data points (we suggest at least over a hundred) such that the results in our theorems are reasonable. In all simulations and real data analysis in Section 6, the period $[0, 1]$ represents a 5-day interval, and we use 1 day as the length of a partition (with typically hundreds of data points in each day), with very good results. From our experience, reducing the length of each partition (provided it still has enough data points, say over a hundred) usually improve the overall performance of the estimator. See Section 5 for more details on how to choose tuning parameters. In the following section, important assumptions in our paper are discussed before we present and explain the main theoretical results.

4 Asymptotic Theory

We introduce some notations first. Let the log-price series $\{\mathbf{X}_t\}$ be adapted to the filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{0 \leq t \leq 1}, \mathbb{P})$. We use $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ to denote the minimum and maximum eigenvalue of a square matrix respectively. Also, if no ambiguity arise, for $j = 1, \dots, L$ we denote $v_s = v_s^j$ which is the s th all-refresh time within partition j , and define

$$\mathcal{F}_{-j} = \mathcal{F}_{\tau_{j-1}} \cup \mathcal{F} / \mathcal{F}_{\tau_j}, \quad \mathcal{F}_s^j = \mathcal{F}_{v_s} / \mathcal{F}_{\tau_{j-1}},$$

with $\mathcal{F}_s^j = \phi$ for $s \leq 0$. They represent the σ -algebra outside of partition j , and that up to time v_s within partition j , respectively. We also define, for $j = 1, \dots, L$,

$$\boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j) = \mathbf{P}_{-j} \text{diag}(\mathbf{P}_{-j}^\top \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) \mathbf{P}_{-j}^\top, \quad (4.1)$$

so that $\boldsymbol{\Sigma}_{\text{Ideal}}(0, 1) = \sum_j \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)$. We first introduce the assumptions for our theorems to hold, followed by some explanations. They are essentially the same as the main assumptions in Lam and Feng (2018). We present here for the sake of completeness and ease of reading for those readers who are interested in these assumptions.

In all assumptions below, we have

$$K = \begin{cases} K_m = N + m, & m = 1, \dots, M, \text{ for the multi-scale method;} \\ 1, & \text{for the kernel method;} \\ 1, & \text{for the pre-averaging method.} \end{cases}$$

Assumptions on the drift $\boldsymbol{\mu}_t$:

(D1) The drift $\boldsymbol{\mu}_t$ has càdlàg components, such that for $s = K, K+1, \dots, n(j)$,

$$\int_{v_{s-K}}^{v_s} \boldsymbol{\mu}_t dt = \mathbf{A}(v_{s-K}, v_s) \mathbf{Z}_{d,s}^j,$$

where $\mathbf{A}(v_{s-K}, v_s) \neq \mathbf{0}$ is a non-random $p \times p$ matrix, has $\|\mathbf{A}(v_{s-K}, v_s)\| = O(p_f^{1/2} K^{1/2} |v_s - v_{s-1}|)$ and can be asymmetric and singular, where $p_f = 1$ if there are no pervasive factors, and $p_f = p$ if there are pervasive factors. Also, $E(\mathbf{Z}_{d,s}^j | \mathcal{F}_{-j}) = \mathbf{0}$ and $\text{var}(\mathbf{Z}_{d,s}^j | \mathcal{F}_{-j}) = \mathbf{I}_p$ almost surely. The random vector $\mathbf{Z}_{d,s}^j \in \mathcal{F}_s^j$ has components conditionally independent of each other given $\mathcal{F}_{-j}$, with eighth moments exist. The drift $\boldsymbol{\mu}_t$ can also be non-random when $\mathbf{Z}_{d,s}^j = (1, 0, \dots, 0)^T$ for all s .

(D2) Write $\mathbf{P}_{-j} = (\mathbf{p}_{1j}, \dots, \mathbf{p}_{pj})$. We assume for each $i = 1, \dots, p$, and $s = rK + q$ for $r = 1, \dots, |S^j(K)|_K$ and $q = 0, 1, \dots, K-1$, there exists $\rho_{d,K,q}^j \in \mathcal{F}_{-j}$ such that $0 \leq \rho_{d,K,q}^j \leq \xi < 1$ with ξ a constant, and for $\ell = K + q, 2K + q, \dots, rK + q$,

$$\begin{aligned} & E((\mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{Z}_{d,\ell}^j)^2 | \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j) \\ &= \rho_{d,K,q}^j (\mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{Z}_{d,\ell-K}^j)^2 + (1 - \rho_{d,K,q}^j) \mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{A}(v_{s-K}, v_s)^\top \mathbf{p}_{ij} + e_{d,\ell-K}^{ij}, \end{aligned}$$

where we define $\mathbf{Z}_{d,\ell}^j \mathbf{Z}_{d,\ell}^{j\top} = \mathbf{I}_p$ and $e_{d,\ell}^{ij} = 0$ for $\ell \leq 0$. The process $\{e_{d,\ell}^{ij}\}$ with $e_{d,\ell}^{ij} \in \mathcal{F}_\ell^j$ has $E(e_{d,\ell}^{ij} | \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j) = 0$ almost surely, and $e_{d,\ell}^{ij} | \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j = O_P(\|\mathbf{A}(v_{s-K}, v_s)\|^2)$.

(D3) Let $\varphi(x) = e^{x^2} - 1$. We assume that for $\ell = 0, 1, \dots, s$,

$$\begin{aligned} & E\left\{\varphi\left(\frac{|\mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{Z}_{d,\ell}^j|^2 - \mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{A}(v_{s-K}, v_s)^\top \mathbf{p}_{ij}|}{(\mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{Z}_{d,\ell-K}^j)^2}\right) \middle| \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j\right\} < \infty, \\ & E\left\{\varphi\left(\frac{|e_{d,\ell}^{ij}|}{(\mathbf{p}_{ij}^\top \mathbf{A}(v_{s-K}, v_s) \mathbf{Z}_{d,\ell-K}^j)^2}\right) \middle| \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j\right\} < \infty. \end{aligned}$$

Assumptions on the volatility $\boldsymbol{\sigma}_t$ and Brownian motion $\mathbf{W}_t$:

(V1) The volatility $\boldsymbol{\sigma}_t$ has càdlàg components, and the Brownian motion $\{\mathbf{W}_t\}$ can be correlated with $\{\boldsymbol{\mu}_t\}$ in general. Write

$$\int_{v_{s-K}}^{v_s} \boldsymbol{\sigma}_t d\mathbf{W}_t = \boldsymbol{\Sigma}(v_{s-K}, v_s)^{1/2} \mathbf{Z}_{v,s}^j,$$

where $\boldsymbol{\Sigma}(v_{s-K}, v_s)$ is a symmetric positive definite $p \times p$ matrix which can be random, with

$$\lambda_{\min}(\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)) \geq C(\tau_{j-1} - \tau_j)^{-1}, \quad \lambda_{\max}(\boldsymbol{\Sigma}(v_{s-K}, v_s)) \asymp \|\mathbf{A}(v_{s-K}, v_s)\|^2 / |v_s - v_{s-K}|,$$

where $C > 0$ is a constant. The process $\{\sigma_t\}$ is independent of all other processes.

Also, $E(\mathbf{Z}_{v,s}^j | \mathcal{F}_{-j}) = \mathbf{0}$ and $\text{var}(\mathbf{Z}_{v,s}^j | \mathcal{F}_{-j}) = \mathbf{I}_p$ almost surely. The random vector $\mathbf{Z}_{v,s}^j \in \mathcal{F}_s^j$ has components conditionally independent of each other given $\mathcal{F}_{-j}$, with eighth moments exist. The $\mathbf{Z}_{v,s}^j$'s are independent of each other for a fixed j .

(V2) Parallel to (D2), but expectations are taken conditional on $\mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j \cup \mathcal{F}_{v_s}^\sigma$, where $\mathcal{F}_t^\sigma$ is the σ -algebra generated by the process $\{\sigma_t\}$ up to time t .

Also, $\rho_{d,K,q}^j$ is replaced by 0, $\mathbf{A}(v_{s-K}, v_s)$ by $\Sigma(v_{s-K}, v_s)^{1/2}$, and $e_{d,\ell}^{ij}$ by 0.

(V3) Parallel to (D3), replacements the same as in (V2).

Assumptions on the microstructure noise ϵ_t :

(E1) Within the j th partition, $E(\epsilon(s)\epsilon(s)^\top | \mathcal{F}_{-j}) = \Sigma_{\epsilon,s}^j$, which is random and independent of all other processes given $\mathcal{F}_{-j}$. Also, $E(\Sigma_{\epsilon,s}^j) = \Sigma_\epsilon^j$, and $\|\Sigma_{\epsilon,s}^j\| \leq \lambda_\epsilon < \infty$ uniformly as $n, p \rightarrow \infty$ where λ_ϵ is a constant. The series $\{\Sigma_{\epsilon,s}^j\}_s$ also satisfies a smoothness condition that $\|\Sigma_{\epsilon,s+l}^j - \Sigma_{\epsilon,s}^j\| \leq |l|/n$ for each $j = 1, \dots, L$.

(E2) Within the j th partition, we can write $\epsilon(s) = (\Sigma_{\epsilon,s}^j)^{1/2} \mathbf{Z}_{\epsilon,s}^j$, with $\mathbf{Z}_{\epsilon,s}^j \in \mathcal{F}_s^j$ having conditionally independent components given $\mathcal{F}_{-j}$. Also $E(\mathbf{Z}_{\epsilon,s}^j | \mathcal{F}_{-j}) = \mathbf{0}$ almost surely and eighth order moments exist for the components of $\mathbf{Z}_{\epsilon,s}^j$.

(E3) Let $\mathcal{F}_t^X$ be the σ -algebra generated by the log-price process up to time t , and $\mathcal{F}_t^\epsilon$ the one by the microstructure noise process up to time t , so that $\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s^X \otimes \mathcal{F}_s^\epsilon$. Then for s_1, s_2 time points within partition j , given $\mathcal{F}_{-j}$, we assume the φ -mixing coefficient between two σ -algebras satisfies

$$\varphi(\mathcal{F}_{s_1}^X, \mathcal{F}_{s_2}^\epsilon | \mathcal{F}_{-j}) = O(n^{-1}) = \varphi(\mathcal{F}_{s_2}^\epsilon, \mathcal{F}_{s_1}^X | \mathcal{F}_{-j}).$$

Also, for $s_2 > s_1$ time points within partition j , we assume

$$\varphi(\mathcal{F}_{s_1}^\epsilon, \mathcal{F}_{s_2}^\epsilon / \mathcal{F}_{s_1}^\epsilon | \mathcal{F}_{-j}) = O(n^{-1}) = \varphi(\mathcal{F}_{s_2}^\epsilon / \mathcal{F}_{s_1}^\epsilon, \mathcal{F}_{s_1}^\epsilon | \mathcal{F}_{-j}).$$

Other assumptions:

(A1) The observation times are independent of $\mathbf{X}(\cdot)$ and $\epsilon(\cdot)$, and the partition boundaries τ_ℓ , $\ell = 0, 1, \dots, L$, satisfy $0 < C_1 \leq \min_{\ell=1, \dots, L} L(\tau_\ell - \tau_{\ell-1}) \leq \max_{\ell=1, \dots, L} L(\tau_\ell - \tau_{\ell-1}) \leq C_2 < \infty$, where C_1, C_2 are generic constants. Also, the all-refresh times v_s , $s = 1, \dots, nL$ satisfy $\max_{s=1, \dots, nL} nL(v_s -$

$v_{s-1}) \leq C_3$ for a generic constant $C_3 > 0$. Moreover, $\max_{\ell=1,\dots,L} nL(\tau_\ell - v_{n(\ell)}) = o(1)$. The sample size in the j th partition has $n(j)/n \rightarrow 1$.

(A2) The pervasive factors, if any, persist within an interval $(v_{s-1}, v_s]$ for $s = 1, \dots, nL$.

There are three more assumptions (A3), (A4) and (A5). They involve the drift and volatility in between the all-refresh and previous-tick times, and are in many ways parallel to assumptions (D1) to (D3) and (V1) to (V3). We present them in Section 7 to aid the flow of the paper. Assumptions (D1) to (D3) and (V1) to (V3) are mainly for the application of a version of Hoeffding's inequality on the sums of martingale differences in Theorem 2.2 of van de Geer (2002), and are also used in Lam and Feng (2018).

The matrix $\mathbf{A}(v_{s-K}, v_s)$ in assumptions (D1) to (D3) can be treated as a factor loading matrix in a factor model when $\boldsymbol{\mu}_t$ is random. The loading matrix $\mathbf{A}(v_{s-K}, v_s)$ is diagonal when the contribution of drift among all assets over v_{s-K} to v_s are conditionally independent given $\mathcal{F}_{-j}$. If there exists a factor structure with r factors where $r \ll p$, the loading matrix $\mathbf{A}(v_{s-K}, v_s)$ becomes a singular matrix whose rank is r . The first r singular values of $\mathbf{A}(v_{s-K}, v_s)$ are then of order $p^{1/2}K^{1/2}|v_s - v_{s-1}|$, with $K^{1/2}|v_s - v_{s-1}|$ accounting for the length of the time interval considered.

Also as introduced in Lam and Feng (2018), Assumption (D2) is about the serial drift dependence quantified by $\rho_{d,K,q}^j < 1$ given $\mathcal{F}_{-j}$. Assumption (D3) says that quadratic forms not too far in time apart can be very different but with sub-Gaussian-tailed probability. When $\boldsymbol{\mu}_t$ is non-random, the matrix $\mathbf{A}(v_{s-K}, v_s)$ can be set as zero except for a non-zero known vector in its first column. For more details please see the corresponding explanations in Lam and Feng (2018).

The assumptions for volatility from (V1) to (V3) have similar forms and meaning, and are parallel to the drift assumptions. One major difference from Assumption (V1) to (V3) in Lam and Feng (2018) is that we use the fact that the $\{\mathbf{Z}_{v,s}^j\}$'s are conditionally independent given $\mathcal{F}_{-j}$, which is in fact a consequence of independent increment in the Brownian motion $\{\mathbf{B}_t\}$. With this, Assumption (V2) certainly becomes

$$E((\mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(v_{s-K}, v_s)^{1/2} \mathbf{Z}_{v,\ell}^j)^2 | \mathcal{F}_{-j} \cup \mathcal{F}_{\ell-K}^j) = \mathbf{p}_{ij}^\top \boldsymbol{\Sigma}(v_{s-K}, v_s) \mathbf{p}_{ij},$$

meaning that $\rho_{d,K,q}^j$ is replaced by 0, $e_{d,\ell}^{ij}$ by 0 and $\mathbf{A}(\cdot, \cdot)$ replaced by $\boldsymbol{\Sigma}(\cdot, \cdot)^{1/2}$. Also, note that Assumption (V1) actually ties the rate of the maximum eigenvalue of $\boldsymbol{\Sigma}(v_{s-K}, v_s)$ to $\|\mathbf{A}(v_{s-K}, v_s)\|$, meaning that if there are pervasive factors such that it spikes the singular value of $\mathbf{A}(\cdot, \cdot)$ by a factor of $p^{1/2}$, then the same factors also spikes the maximum eigenvalues of $\boldsymbol{\Sigma}(\cdot, \cdot)$. It makes sense since pervasive factors should have effects on both the drift and the volatility.

Assumption (E1) allows for time-varying covariance matrix for the microstructure noise. A smoothness

condition for the time-varying covariance matrix is needed for the bias-corrected pre-averaging method to successfully remove the residual covariance effects from the microstructure noise. Assumption (E3) particularly assumes a weak dependence between the log-price process and the microstructure noise process within partition j , as well as a weak serial dependence among the microstructure noise vectors, when $\mathcal{F}_{-j}$ is given. This assumption is inspired by Chen and Mykland (2017), where they assumed that given the entire information of the log-price process, the microstructure noise at different time points are independent. In our case, we are not given the entire picture of the log-price process, but not far from that either since with $\mathcal{F}_{-j}$ we are given $nL - n(j)$ data points from the total of nL . Then instead of assuming the microstructure noise vectors are independent, we assume that they are weakly dependent, and with n larger (i.e., more information at more time points is available outside partition j), the dependence is weaker.

The first part of Assumption (A1) is automatically satisfied if the boundary set $\{\tau_\ell\}_{0 \leq \ell \leq L}$ is pre-set, for instance, to be the daily opening or closing time of the L days of data, or a quarter of it. See also Section 5 on a criterion in choosing these tuning parameters. Assumption (A2) means that the pervasive factors are either present between two all-refresh times, or they are absent.

Theorem 1. *Let Assumptions (D1) to (D3), (V1) to (V3), (E1) to (E3) and (A1) to (A5) hold. Then as $n, p \rightarrow \infty$ such that $p/n \rightarrow c \geq 0$,*

$$\max_{j=1, \dots, L} \left\| \widehat{\Sigma}(\tau_{j-1}, \tau_j)^M \Sigma_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p \right\| = O_P(n^{-1/6} + p_f^{1/2} n^{-1/2}) = \left\| \widehat{\Sigma}(0, 1)^M \Sigma_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p \right\|,$$

where $\|\cdot\|$ denotes the spectral norm of a matrix, and $p_f = 1$ if there are no factors, while $p_f = p$ if there are pervasive factors in the log-price processes.

See the supplementary materials of the paper for the proof. It is known that the rate of convergence for univariate realized volatility estimator in Zhang (2006) or the rate for multi-scale volatility matrix estimator under sparse assumption in Kim et al. (2016) is $n^{-1/4}$, the best achievable one. For our NER-MSRVM estimator, we have the slower rate $n^{-1/6}$ when there are no pervasive factors because we are not assuming sparsity of $\Sigma(\tau_{j-1}, \tau_j)$, although we still have the same rate as NERIVE in Lam and Feng (2018). In (3.4), the scales we use are $K_m = m + N$, $m = 1, \dots, M$. While Zhang (2006) and Kim et al. (2016) use $N \asymp n^{1/2}$ with $M \asymp n^{1/2}$, we use $N \asymp n^{2/3}$ and $M \asymp n^{1/2}$, so that $K_m \asymp n^{2/3}$, the same magnitude as the scale used in Zhang (2011) and Lam and Feng (2018) in their two-scale estimator. We find that this larger scale is needed to remove the adverse effects of large p without sparsity while removing the bias from microstructure noise.

When there are pervasive factors, $p_f = p$, and the rate tell us that we need $p = o(n)$ in order to still have guaranteed convergence as $n, p \rightarrow \infty$. What happens is that some drift terms can somehow dominates

the volatility terms when there are pervasive factors which spike up the maximum singular value of $\mathbf{A}(\cdot, \cdot)$ and $\mathbf{\Sigma}(\cdot, \cdot)^{1/2}$ simultaneously by an order of $p^{1/2}$. Ultimately, when $p_f = p$ which has the same order as $n^{2/3}$, then we still have $n^{-1/6}$ as the rate of convergence, which coincides with the results for NERIVE in Lam and Feng (2018). If the factors are not pervasive, then $p_f = p^{1-\delta}$ for some $0 < \delta < 1$ which represents the strength of the factors (Lam and Yao, 2012, Lam et al., 2011). Clearly we can still have $p/n \rightarrow c > 0$ in such a case, and can retain the rate $n^{-1/6}$ if $\delta \geq 1/3$.

In practice, a larger M means we are using more scales, which can improve the estimation in general. This aligns with our simulation results which show that NER-MSRVM performs better than NERIVE in Lam and Feng (2018) which uses only two scales.

Theorem 2. *With the same conditions as in Theorem 1, as $n, p \rightarrow \infty$ such that $p/n \rightarrow c \geq 0$, we have*

$$\max_{j=1, \dots, L} \|\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^K \mathbf{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p\| = O_P(p_f^{1/2} n^{-1/4}) = \|\widehat{\mathbf{\Sigma}}(0, 1)^K \mathbf{\Sigma}_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p\|,$$

with $H \asymp n^{1/2}$ and $J \asymp p_f^{-1/2} n^{1/4}$, where $p_f = 1$ if there are no factors, and $p_f = p$ if there are pervasive factors. For the positive semi-definite version, we have

$$\max_{j=1, \dots, L} \|\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^{KP} \mathbf{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p\| = O_P(p_f^{2/5} n^{-1/5}) = \|\widehat{\mathbf{\Sigma}}(0, 1)^{KP} \mathbf{\Sigma}_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p\|,$$

with $H \asymp p_f^{-1/5} n^{3/5}$ and $J \asymp p_f^{-2/5} n^{1/5}$ respectively.

See the supplementary materials of the paper for the proof. Kim et al. (2016) introduces KRVM which has a faster rate of convergence at $n^{-1/4}$, and KRPVM, which has rate only at $n^{-1/5}$ but is guaranteed to be positive semi-definite. The above theorem shows that NER-KRVM is approaching $\mathbf{\Sigma}_{\text{Ideal}}(0, 1)$ in spectral norm at a rate of $n^{-1/4}$ when there are no factors, and hence it is positive definite in probability. The same goes for NER-KRPVM, but the rate is slower at $n^{-1/5}$. When there are pervasive factors in the log-price processes, all rates of convergence to the ideal estimator are slower. As explained after Theorem 1, this happens since some terms related to the drift can dominate when there are pervasive factors. This theorem suggests that for our regularized estimator for the kernel method, the bias-corrected version NER-KRVM is always better than NER-KRPVM since they are both positive definite in probability anyway, while the former converges faster to the ideal estimator $\mathbf{\Sigma}_{\text{Ideal}}(0, 1)$. This is true as long as $p = o(n^{1/2})$, when $p^{2/5} n^{-1/5}$ is going to 0 slower than $p^{1/2} n^{-1/4}$.

Theorem 3. *With the same conditions as in Theorem 1, and $n, p \rightarrow \infty$ such that $p/n \rightarrow c \geq 0$,*

$$\max_{j=1, \dots, L} \|\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)^P \mathbf{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p\| = O_P(p_f^{3/8} n^{-1/4}) = \|\widehat{\mathbf{\Sigma}}(0, 1)^P \mathbf{\Sigma}_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p\|,$$

with $Q \asymp p_f^{-1/4} n^{1/2}$, where $p_f = 1$ if there are no factors, and $p_f = p$ if there are pervasive factors. We also have

$$\max_{j=1,\dots,L} \|\widehat{\Sigma}(\tau_{j-1}, \tau_j)^{PP} \Sigma_{\text{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p\| = O_P(p_f^{2/5} n^{-1/5}) = \|\widehat{\Sigma}(0, 1)^{PP} \Sigma_{\text{Ideal}}(0, 1)^{-1} - \mathbf{I}_p\|,$$

with $Q \asymp p_f^{-1/5} n^{3/5}$.

See the supplementary materials of the paper for the proof. Same as kernel estimators, Kim et al. (2016) also gives two versions of pre-averaging estimators: the bias-corrected version PRVM, which is not guaranteed to be positive semi-definite but is converging at a rate of $n^{-1/4}$, and the positive semi-definite version PRPVM, which has a slower rate of convergence at $n^{-1/5}$. Our results show that the regularized estimators are all positive definite in probability, and when there are no factors, the rate of convergence to the ideal estimator $\Sigma_{\text{Ideal}}(0, 1)$ for NER-PRVM is $n^{-1/4}$. For NER-PRPVM, it is $n^{-1/5}$ under $p/n \rightarrow c \geq 0$. These rates are the same as those for the non-regularized estimators with finite p . Same as Theorem 1 and 2, some terms related to the drift can dominate as $p \rightarrow \infty$, and hence ultimately all rates of convergence are slower. However, NER-PRVM still has a faster rate of convergence to the ideal estimator at $p^{3/8} n^{-1/4}$ when compared to that for NER-PRPVM at $p^{2/5} n^{-1/5}$. Note also that $p = o(n^{2/3})$ is needed for NER-PPVM while $p = o(n^{1/2})$ is needed for NER-PRPVM for guaranteed convergence when there are pervasive factors.

Comparing the rate $p_f^{3/8} n^{-1/4}$ for NER-PRVM to $p_f^{1/2} n^{-1/4}$ for NER-KRVM, it is clear that when there are pervasive factors, the pre-averaging estimator can converge a bit faster to the ideal estimator. It also allows $p = o(n^{2/3})$ compared to only $p = o(n^{1/2})$ allowed for NER-KRVM. The practical performance for NER-PRVM is also better than NER-KRVM in many scenarios as demonstrated in Section 6.1. The simulation results also highlight that NER-PRVM is better than NER-PRPVM in general. All our regularized estimators constructed during simulations and real data analysis are also positive definite, showing that the convergence is in fact pretty quick. In the following section, we also show that our regularized estimators are adaptive to jumps removal.

4.1 Removal of jumps

We consider a jump-diffusion log-price process

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \boldsymbol{\sigma}_t d\mathbf{W}_t + d\mathbf{J}_t, \quad t \in [0, 1], \quad (4.2)$$

where $\boldsymbol{\mu}_t$ and $\boldsymbol{\sigma}_t$ are the same as the pure diffusion model (2.1), and $\mathbf{J}_t = (J_t^{(1)}, \dots, J_t^{(p)})^\top$ is a p -dimensional right-continuous pure jump process. For the j th asset,

$$J_t^{(j)} = \sum_{\ell=1}^{N_t^{(j)}} B_\ell^{(j)}, \quad t \in [0, 1],$$

where each count process $N_t^{(j)}$ is assumed to be finite and can be correlated with each other. The same holds true for each jump size $B_\ell^{(j)}$. The quadratic covariation over $[0, 1]$ for the process $\mathbf{X}_t$ is denoted by

$$QV = \int_0^1 \boldsymbol{\sigma}_t \boldsymbol{\sigma}_t^\top dt + \sum_{0 \leq t \leq 1} \Delta \mathbf{J}_t \Delta \mathbf{J}_t^\top, \quad (4.3)$$

where $\Delta \mathbf{J}_t = \mathbf{J}_t - \mathbf{J}_{t-}$. An off-diagonal entry in $\Delta \mathbf{J}_t \Delta \mathbf{J}_t^\top$ will only be non-zero when both the corresponding log-price processes have jumps at the same time (cojumps) at least once. This can happen when market news affect some stocks at the same time. If jumps occur regularly, we should account for jump risks by considering QV as a whole rather than just the integrated volatility matrix.

For estimating the integrated covariance matrix, same as Lam and Feng (2018), we propose to use the wavelet method introduced in Fan and Wang (2007) to estimate $\mathbf{J}_t$ by $\hat{\mathbf{J}}_t$. We then remove the jumps in $\mathbf{X}_t$ and construct an integrated volatility matrix estimator $\hat{\boldsymbol{\Sigma}}(0, 1)$ using the jump-removed data $\mathbf{Y}^*(s) = \mathbf{Y}(s) - \hat{\mathbf{J}}_{v_s}$, $s = 1, \dots, nL$. An estimator of QV is then

$$\widehat{QV} = \hat{\boldsymbol{\Sigma}}(0, 1) + \sum_{0 \leq t \leq 1} \Delta \hat{\mathbf{J}}_t \Delta \hat{\mathbf{J}}_t^\top.$$

To apply this wavelet method successfully, more assumptions are need:

(W1) The wavelets used in jump estimation are differentiable.

(W2) For the jump part of $X_t^{(j)}$ in $[0, 1]$ for $j = 1, \dots, p$, its jump locations $\eta_\ell^{(j)}$ and jump sizes $B_\ell^{(j)}$ satisfy

$$N_1^{(j)} < \infty, \eta_1^{(j)} < \dots < \eta_\ell^{(j)} < \dots, 0 < |B_\ell^{(j)}| < \infty \text{ almost surely.}$$

(W3) The number of stocks involved in a cojump is finite.

Assumptions (W1) and (W2) are technical assumptions adapted from Fan and Wang (2007). Assumption (W2) means that we are dealing with finite number of jumps for each log-price process, and the size of each jump is bounded almost surely. If Assumption (W3) is not satisfied, then the rate of convergence of $\hat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ in Theorem 1 and 3 using the jumps-removed data will be dependent on how many stocks is

involved in a cojump. Our assumptions allow the jump process to be dependent on the drift, volatility and the microstructure noise process in general. Readers who are interested in the details in how jumps are detected and estimated are referred to Fan and Wang (2007) and Lam and Feng (2018) for more details.

Theorem 4. *Let all the assumptions in Theorem 3 hold, as well as (W1) to (W3) for the jump-diffusion model (4.2). Using the jumps-removed all-refresh log-price data $\mathbf{Y}^*(s) = \mathbf{Y}(s) - \hat{\mathbf{J}}_{v_s}$, $s = 1, \dots, nL$ in constructing the integrated covariance matrix estimator in (3.13), (3.14) or (3.15), the same conclusions in Theorem 1, 2 and 3 hold. Moreover, we have*

$$\left\| \sum_{0 \leq t \leq 1} (\Delta \mathbf{J}_t \Delta \mathbf{J}_t^\top - \Delta \hat{\mathbf{J}}_t \Delta \hat{\mathbf{J}}_t^\top) \right\| = O_P(n^{-1/4}).$$

5 Practical Implementation

For the nonparametric eigenvalue regularization introduced in Section 3.4, there is one common feature that can be tuned, namely the partition $(\tau_{j-1}, \tau_j]$ of the period $[0, 1]$ (thus also determining L itself which represents the number of partitions). For MSRVM in (3.4), we need to tune the scale parameters M and N . For KRVM and KRPVM in (3.6) and (3.7) respectively, we need to tune the window length H and the jittering length J , while PRVM and PRPVM in (3.9) and (3.10) respectively need the average range Q to be determined.

Since M , N for MSRVM, H , J for KRVM and KRPVM, and Q for PRVM and PRPVM all have suggested order of magnitudes in Section 3.1, 3.2 and 3.3 respectively, we use exactly the value of their order with proportionality constant equals 1. It turns out that for all methods, their performances are not really sensitive to the proportionality constants. Incidentally, Lam and Feng (2018) uses 1 as proportionality constant as well for TSRVM with their scale parameter $[n^{2/3}]$.

The more important tuning parameter is in fact the number of partition L . Same as Lam and Feng (2018), we propose a criterion as follows:

$$g(\boldsymbol{\tau}) = \left\| \sum_{j=1}^L \left(\hat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j) - \tilde{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j) \right) \right\|_F^2, \quad (5.1)$$

where $\tilde{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ is the non-regularized estimator and $\hat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ is the nonparametric eigenvalue regularized version. Bickel and Levina (2008) proposes this function to choose the banding number in a large covariance matrix estimator which is banded. Lam and Feng (2018) suggests dividing the time interval into equal length partitions, provided each partition has enough number of data points, typically in hundreds or at the order of p , whichever is the largest. In practice this works well for all estimators. In Section 6 we

use a 1-day partition length for a 5-day time period with good results.

6 Empirical Results

6.1 Simulations

We generate 100 trading days of log-price data, where the price processes and the asynchronous transaction times are simulated independently. The observed log-price is defined as $X_t^{o(i)} = X_t^{(i)} + \epsilon_t^{(i)}$, where $X_t^{(i)}$ is the i th asset's latent log-price, $i = 1, \dots, p$, and $\epsilon_t^{(i)} \stackrel{iid}{\sim} N(0, 0.0005^2)$. We use a Heston-like multivariate factor model with stochastic volatilities used in Lam and Feng (2018) to generate $\mathbf{X}_t$ as follows:

$$dX_t^{(i)} = \mu^{(i)} dt + \sqrt{1 - (\rho^{(i)})^2} \sigma_t^{(i)} dB_t^{(i)} + \rho^{(i)} \sigma_t^{(i)} dW_t + C \nu^{(i)} dZ_t, \quad i = 1, \dots, p, \quad (6.2)$$

where $\{W_t\}$, $\{Z_t\}$ and the $\{B_t^{(i)}\}$'s are independent standard Brownian motions. The processes $\{W_t\}$ and $\{Z_t\}$ imitate factors in the market. The constant $C = 1_{\{\text{model 2}\}}$ is 0 for the first model we consider. We set $\rho^{(i)} = -0.7C$, so that it is 0 in the first model, and hence there are no factors. For the second model, $C = 1$, so that it contains two factors. The spot volatility $\sigma_t^{(i)} = \sqrt{\varrho_t^{(i)}}$ has $\varrho_t^{(i)}$ follows the Cox-Ingersoll-Ross (CIR) process

$$d\varrho_t^{(i)} = \kappa^{(i)}(\theta^{(i)} - \varrho_t^{(i)})dt + \xi^{(i)} dU_t^{(i)},$$

where the $\{U_t^{(i)}\}$'s are independent standard Brownian motions. Other parameters of $X_t^{(i)}$ are set at $(\mu^{(i)}, \kappa^{(i)}, \xi^{(i)}, \theta^{(i)}) = (0.03x_1^{(i)}, 1.1x_2^{(i)}, 0.5x_3^{(i)}, 0.25x_4^{(i)})$ and $\nu^{(i)} = \sqrt{\theta^{(i)}}$, where the $x_j^{(i)}$'s are independent uniform random variables on the interval $[0.7, 1.3]$. The initial value of each log-price $X_0^{(i)}$ is set randomly on the interval $[0.5, 1.5]$ and the starting spot volatility $\varrho_0^{(i)}$ on the interval $[0.5\theta^{(i)}, 1.5\theta^{(i)}]$.

For the transaction times, we generate p independent Poisson processes with intensities $\lambda_1, \dots, \lambda_p$ respectively. To examine the finite sample performance for our proposed methods, we generate data using 23400 seconds for a full trading day. Each λ_i is set to be $0.01i \times 23400$, where $i = 1, \dots, p$.

To gauge the performance of each method, we use the Frobenius error norm,

$$\text{Error} = \text{tr}(\widehat{\Sigma}(0, 1) - \Sigma(0, 1)),$$

where $\Sigma(0, 1)$ is evaluated using the simulated latent log-prices at the finest grid (1 per second). We divide the 100 trading days into disjoint 5-day intervals, and for each method we calculate the error above over each 5-day interval.

We compare four different methods and their nonparametric eigenvalue regularized versions. The first method includes TSRVM and NER-TSRVM, where NER-TSRVM is exactly NERIVE in Lam and Feng (2018). The second one includes MSRVm and NER-MSRVm. As discussed in Section 3.1, we use $N = n^{2/3}$ rather than $n^{1/2}$ (with proportionality constant exactly 1, as discussed in Section 5), and $M = \lceil n^{1/2} \rceil$ where $\lceil a \rceil$ represents the integer closest to a . We also denote by mMSRVm and NER-mMSRVm the versions with $N = n^{1/2}$. We compare both versions in order to demonstrate the need for updating $N = n^{1/2}$ to $N = n^{2/3}$ when no sparsity is assumed. The third method includes KRVm, NER-KRVm and NER-KRPVm, while the fourth one includes PRVm, NER-PRVm and NER-PRPVm.

We set $p = 100$ for all simulations relating to Figure 1 to 9. Figure 1 shows the results where the data has no jumps (upper plot), and with 3 jumps in each log-price process (middle plot) where all jumps are i.i.d. $N(0, 1/30)$ and processed according to Section 4.1. We can see that all regularized estimators are performing better than their respective non-regularized versions, with the most pronounced differences from the kernel and the pre-averaging methods. NER-PRVm has the best performance, followed closely by PRVm itself. This does not stay true however when there are less data points. The bottom plot of Figure 1 shows the results when there are only 5000 seconds per day, equivalent to having around 4 times less data points. While all methods perform worse, we can see all regularized estimators perform much better than their non-regularized counterparts. PRVm is in fact the worst performing method now, with huge variability. However, NER-PRVm is still the best performer, showing our regularization works well, and is needed even more so when there are less data points.

In Figure 2, we compare NER-MSRVm ($N = n^{2/3}$) and NER-mMSRVm ($N = n^{1/2}$). Clearly, NER-MSRVm is much better than NER-mMSRVm in performance, showing that without sparsity assumption it is better to use larger scales of order $n^{2/3}$, as depicted in Theorem 1.

Figure 3 shows that NER-KRVm and NER-PRVm are both far better than their respective versions constructed from positive semi-definite estimators KRPVm and PRPVm. This is expected since the rate of convergence to the ideal estimator is faster for both NER-KRVm and NER-PRVm. In fact, all nonparametric eigenvalue regularized estimators constructed are positive definite in our simulations.

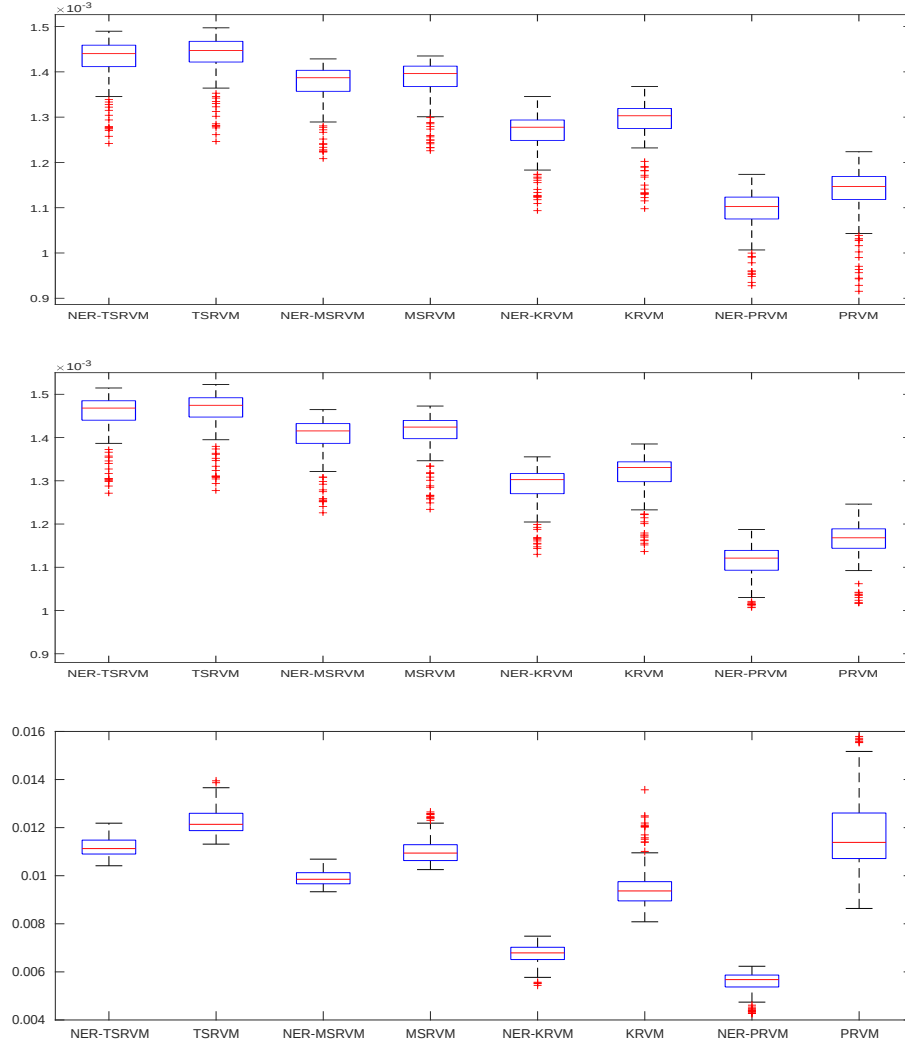


Figure 1: Boxplots of Frobenius errors of NER-TSRVM, TSRVM, NER-MSRVM, MSRVM, NER-KRVM, KRVM, NER-PRVM and PRVM for $C = 0$ (no factors). Upper: No jumps. Middle: Jumps detected and removed with jumps size following i.i.d. $N(0,1/30)$. Lower: No jumps, but a trading day has only 5000 seconds.

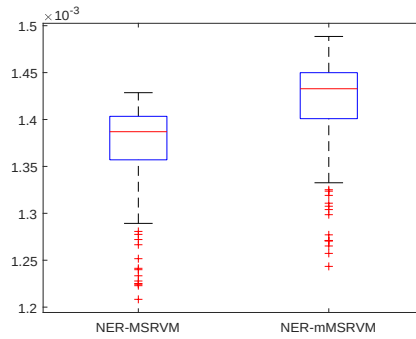


Figure 2: Boxplots of Frobenius errors for NER-MSRVM and NER-mMSRVM, $C = 0$ (no factors) with no jumps.

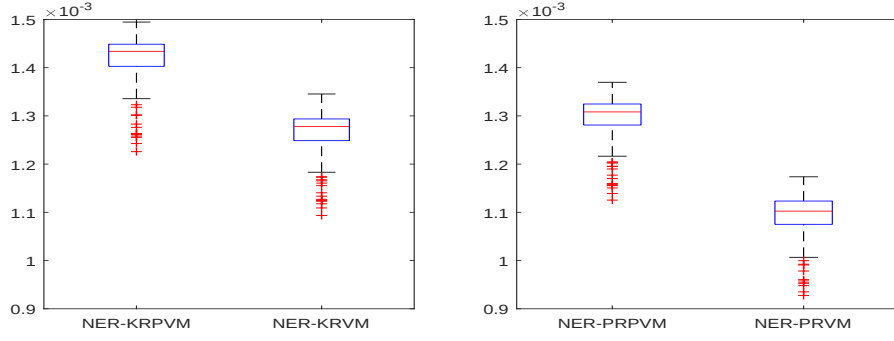


Figure 3: Boxplots of Frobenius errors when $C = 0$ (no factors) with no jumps. Left: NER-KRPVM and NER-KRVM. Right: NER-PRPVM and NER-PRVM.

Dai et al. (2017) proposed a principal orthogonal complement thresholding (POET) method based on pre-averaging data to remove the effects from microstructure noise, and at the same time capture any factors and threshold the residual covariance matrix. We apply this method on our simulated data and denote it as PR-POET. Furthermore, a pure POET method without the pre-averaging step is also included in our simulation. Figure 4 shows the results. It is clear that NER-PRVM outperforms both POET and PR-POET by a large margin.

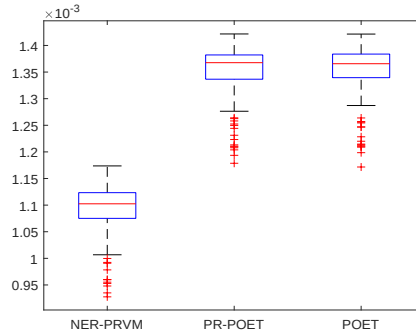


Figure 4: Boxplots of Frobenius errors of NER-PRVM, PR-POET and POET when $C = 0$ (no factors) and no jumps.

Figure 5 to 8 are analogous to Figure 1 to 4 respectively but with $C = 1$, so two factors are included in the log-price vector $\mathbf{X}_t$. We can still see from Figure 5 that all nonparametric eigenvalue regularized estimators outperform their non-regularized counterparts. NER-PRVM is still the best, closely followed by PRVM. The use of $N = n^{2/3}$ still results in a much better NER-MSRVM estimator compared with using $N = n^{1/2}$ as seen in Figure 6, while Figure 7 still demonstrates that NER-KRVM and NER-PRVM are better than NER-KRPVM and NER-PRPVM respectively. Finally, Figure 8 shows that NER-PRVM is still better than PR-POET and POET.

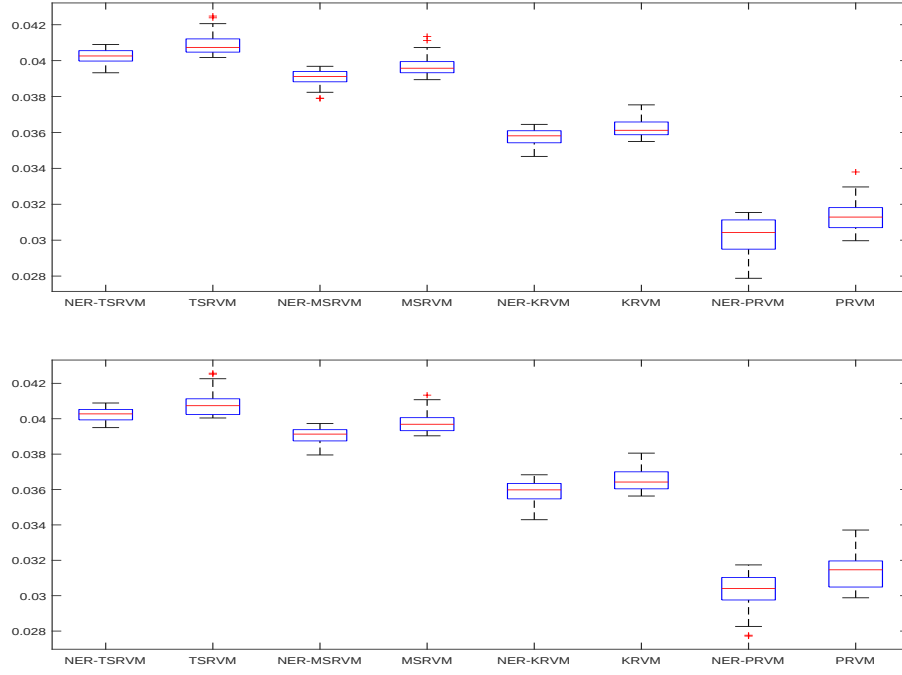


Figure 5: Boxplots of Frobenius errors of NER-TSRVM, TSRVM, NER-MSRVM, MSRVM, NER-KRVM, KRVM, NER-PRVM and PRVM for $C = 1$ (two factors). Upper: No jumps. Lower: Jumps detected and removed with jumps size following i.i.d. $N(0, 1/30)$.

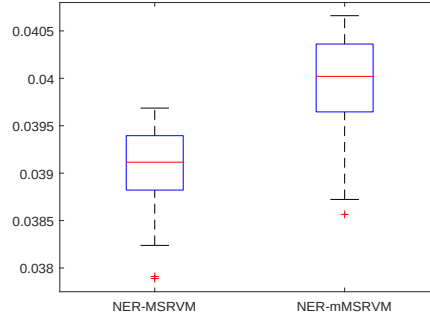


Figure 6: Boxplots of Frobenius errors for NER-MSRVM and NER-mMSRVM, $C = 1$ (two factors) with no jumps.

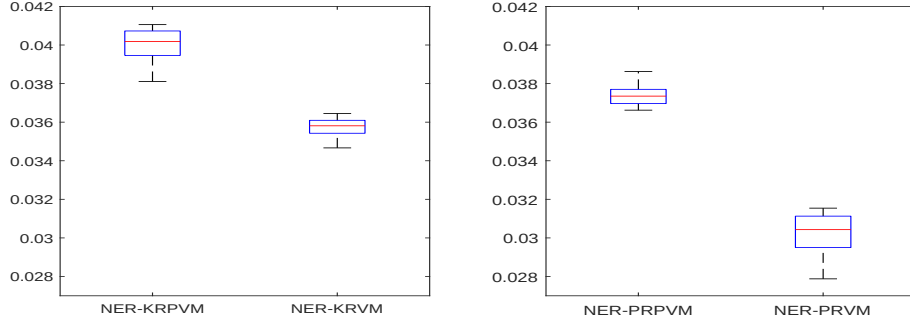


Figure 7: Boxplots of Frobenius errors when $C = 1$ (two factors) with no jumps. Left: NER-KRPVM and NER-KRVM. Right: NER-PRPVM and NER-PRVM.

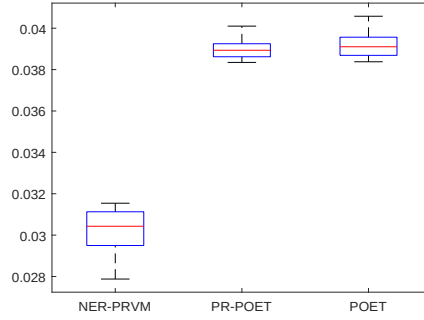


Figure 8: Boxplots of Frobenius errors of NER-PRVM, PR-POET and POET when $C = 1$ (two factors) and no jumps.

All the simulations above assumed a standard deviation of 0.0005 for the microstructure noise. We increase this in steps from 0.0005 to 0.005 to see what happens when the noise level increases. For academic interests we also increase it from 0.01 to 0.1 to see how each method could break down under such unreasonably large noise level. Figure 9 shows that NER-KRVM will eventually be better than NER-PRVM as the noise level increases, with or without factors. However, both of them will be outperformed by NER-MSRVM when the noise level is really high, showing that NER-MSRVM is the most resistant to the level of microstructure noise, which is closely followed by NER-KRVM. NER-TSRVM will still be the worst throughout. It would seem that pre-averaging, which performs smoothing for all data points, suffer the most (apart from NER-TSRVM) under high noise level. NER-KRVM, which only smooths the beginning and the end of the data essentially and find a weighted average of autocovariance matrices, suffer much less. NER-MSRVM, without smoothing the data at all but with a designed weighted average of integrated volatility matrices calculated at different scales, has the best performance. In fact NER-MSRVM has a very consistent performance over a wide range of noise level, showing the method is very resistant to the effect of microstructure noise.

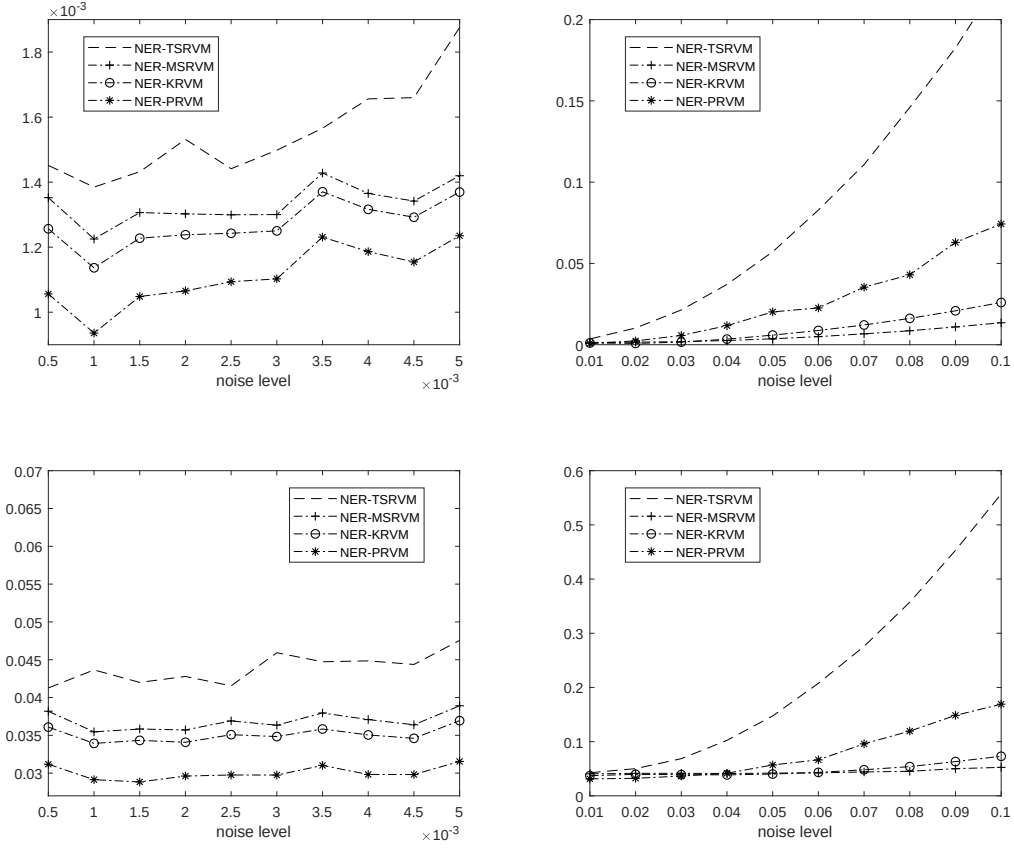


Figure 9: Frobenius errors when we increase the standard deviation of the microstructure noise (no jumps). Upper panel: No factors in model (6.2) ($C = 0$). Lower panel: Two factors in model (6.2) ($C = 1$).

Since we have set $p = 100$ so far, we want to showcase how each method reacts to an increasing p in Figure 10. Clearly NER-PRVM stays the best from $p = 10$ to $p = 100$, with or without factors/jumps in the log-price processes. The performance gap between NER-PRVM and other methods also widens as p increases.

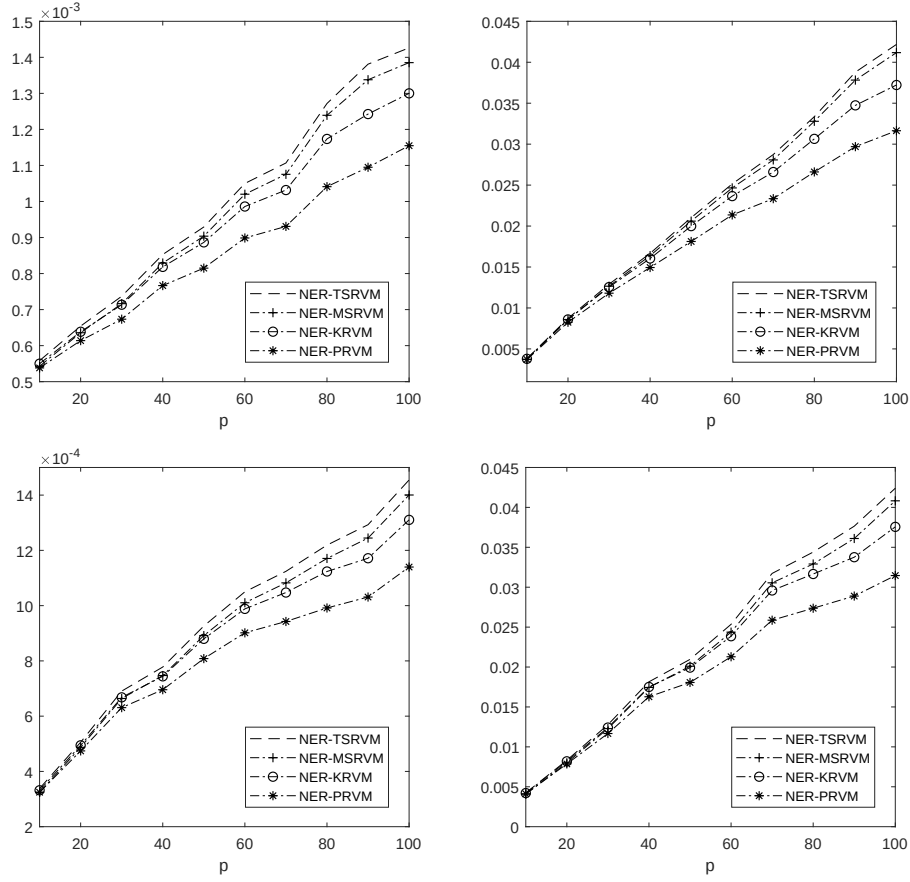


Figure 10: Frobenius errors with increasing p . Left panel: $C = 0$ (no factors). Right panel: $C = 1$ (two factors). Upper panel: No jumps. Lower panel: Jumps detected and removed with jumps size following i.i.d. $N(0,1/30)$.

6.2 Real data analysis

6.2.1 Minimum variance portfolio allocation

We first present a Theorem on minimum variance portfolio using our integrated volatility matrix estimators.

Define $\mathbf{1}_p$ a column vector of p ones, the theoretical minimum variance portfolio has weights defined by

$$\mathbf{w}_{\text{theo}} = \frac{\boldsymbol{\Sigma}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^\top \boldsymbol{\Sigma}(0,1)^{-1}\mathbf{1}_p}.$$

An estimated minimum variance portfolio weight vector is then

$$\hat{\mathbf{w}} = \frac{\hat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^\top \hat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}.$$

We also define the maximum exposure of a portfolio $\mathbf{w}$ as $\|\mathbf{w}\|_{\max} = \max_i |w_i|$.

Theorem 5. *Let all assumptions in Theorem 3 hold. If there are no pervasive factors with $p/n \rightarrow c > 0$, or there are pervasive factors but $p^{3/2}/n \rightarrow c > 0$, the maximum exposures of $\mathbf{w}_{\text{theo}}$ and $\hat{\mathbf{w}}$, with $\hat{\Sigma}(0,1)$ being equal to $\hat{\Sigma}(0,1)^M$, $\hat{\Sigma}(0,1)^K$, $\hat{\Sigma}(0,1)^{KP}$, $\hat{\Sigma}(0,1)^P$ or $\hat{\Sigma}(0,1)^{PP}$, will satisfy in probability*

$$p^{1/2}\|\hat{\mathbf{w}}\|_{\max}, p^{1/2}\|\mathbf{w}_{\text{theo}}\|_{\max} \leq \frac{\max_{1 \leq j \leq L} \lambda_{\max}(\Sigma(\tau_{j-1}, \tau_j))}{\min_{1 \leq j \leq L} \lambda_{\min}(\Sigma(\tau_{j-1}, \tau_j))}.$$

If there are no pervasive factors and $p/n \rightarrow c > 0$, the actual risk of $\hat{\mathbf{w}}$ (the same as above) and $\mathbf{w}_{\text{theo}}$, with an actual risk of $\mathbf{w}$ defined by $R^{1/2}(\mathbf{w}) = (\mathbf{w}^T \Sigma(0,1) \mathbf{w})^{1/2}$, satisfy in probability

$$p^{1/2} R^{1/2}(\hat{\mathbf{w}}) \leq \frac{\max_{1 \leq j \leq L} \lambda_{\max}(\Sigma(\tau_{j-1}, \tau_j))}{\min_{1 \leq j \leq L} \lambda_{\min}(\Sigma(\tau_{j-1}, \tau_j))} \cdot \lambda_{\max}^{1/2}(\Sigma(0,1)),$$

$$p^{1/2} R^{1/2}(\mathbf{w}_{\text{theo}}) \leq \lambda_{\max}^{1/2}(\Sigma(0,1)).$$

If there are pervasive factors and $p^{3/2}/n \rightarrow c > 0$, then $R(\hat{\mathbf{w}}) = O_P(\lambda(\Sigma(0,1))) = O_P(p)$, and the bound for $R(\mathbf{w}_{\text{theo}})$ remains the same as above.

If Assumptions (W1) to (W3) hold also under the jump-diffusion model (4.2), then the same conclusions as above hold, as long as we are using the jumps removal procedure described in Section 4.1.

Theorem 5 is exactly the same as Theorem 5 in Lam and Feng (2018), and the proof is also the same and hence interested readers are encouraged to read the proof of Theorem 5 there. The maximum exposure bound is important since it is clear that the theoretical minimum variance portfolio also satisfies this bound. In practice, unless we are using other methods and building towards a concentrated portfolio, as far as minimum portfolio is concerned, we do not want to invest in a single asset too much in any single period of time, especially when the theoretical portfolio is not doing so.

6.2.2 NYSE data analysis

In this study, we choose the stocks based on two lists, the “Fifty Most Active Stocks on NYSE, Round Lots (mils. of shares), 2013” and “Fifty Most Active Stocks by Dollar Volume on NYSE (\$ in mils.), 2013”, from the New York Stock Exchange Data official website <http://www.nyxdata.com/>. There are 26 stocks appearing in both of the lists above, and 74 stocks in either of them. The data is the same as those used in Lam and Feng (2018). We downloaded all the trading transactions of these 74 stocks in Year 2013 from the Wharton Research Data Services (WRDS, <https://wrds-web.wharton.upenn.edu/>). We omit the stock Sprint Corporation due to missing price data. We first clean all the data by the R-package “highfrequency”, which follows the high frequency data cleaning steps presented in Barndorff-Nielsen et al.

(2009). We conduct our portfolio allocation study on two set of stocks, one with the $p = 26$ stocks appearing in both lists, and the other with $p = 73$ stocks appearing in either of the lists.

The quantities to be compared for different portfolios are as follows. We rebalance our portfolio daily with a 5-day training window. We calculate the annualized portfolio return and annualized out-of-sample standard deviation, given respectively by

$$\hat{\mu} = 250 \times \frac{1}{245} \sum_{i=6}^{250} \mathbf{w}^\top \mathbf{r}_i, \quad \hat{\sigma} = \left(250 \times \frac{1}{245} \sum_{i=6}^{250} (\mathbf{w}^\top \mathbf{r}_i - \frac{\hat{\mu}}{250})^2 \right)^{1/2}.$$

The out-of-sample standard deviation is a good indicator of how much risk is associated with a portfolio (Demiguel and Nogales, 2009), and is our main quantity for performance comparisons.. We also calculate the Sharpe ratio $\hat{\mu}/\hat{\sigma}$. In the face of the results in Theorem 5, the average maximum exposure and the maximum of the maximum exposure over the whole investment horizon are two important measures for comparisons too.

It is clear that minimum variance portfolios using our regularized estimators perform better than their respective non-regularized ones, with NER-PRVM having the smallest our-of-sample standard deviation as well as both the smallest average maximum exposure and the smallest maximum of maximum exposure over the whole investment horizon.

7 Appendix

We present the last set of assumptions which are required for Theorem 1,2,3 and 4 to hold. We first need to decompose $\mathbf{X}_{v_s} - \mathbf{X}(s)$. Consider the previous-tick time $t_s^i \in (v_{s-1}, v_s]$ for the i th asset, which should satisfy

$$v_{s-1} < t_s^{i_1} \leq t_s^{i_2} \leq \dots \leq t_s^{i_p} = v_s,$$

where $\{i_1, \dots, i_p\}$ is some permutation of $1, \dots, p$. Letting b_s (assumed $o(p)$) denote the number of tides, we can write the above as

$$v_{s-1} < t_s^{j_1} < t_s^{j_2} < \dots < t_s^{j_{p-b_s}} = v_s,$$

where $j_1, \dots, j_{p-b_s} \in \{1, \dots, p\}$.

Then we can write, for $s = 1, \dots, nL$,

$$\mathbf{X}_{v_s} - \mathbf{X}(s) = \sum_{m=1}^{p-b_s-1} \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(m+1) + \sum_{m=1}^{p-b_s-1} \mathbf{D}_m^s \Sigma(t_s^{j_m}, t_s^{j_{m+1}})^{1/2} \mathbf{Z}_{v,s}^j(m+1), \quad (7.1)$$

Methods	Out-of-Sample SD (%)	Aver Max Abs Weight(%)	Max Max Abs Weight(%)	Portfolio Return(%)	Sharpe Ratio
$p=26$					
NER-TSRVM	4.7	20 ₍₆₎	44	17.1	3.63
TSRVM	5.7	41 ₍₁₄₎	94	16.3	2.85
NER-MSRVM	4.6	20 ₍₆₎	42	17.8	3.86
MSRVM	5.7	42 ₍₁₄₎	98	17.4	3.05
NER-mMSRVM	4.9	21 ₍₇₎	49	17.7	3.61
mMSRVM	5.7	42 ₍₁₆₎	108	17.4	4.39
NER-KRPVM	4.6	19 ₍₅₎	39	20.2	4.39
KRPVM	5.3	27 ₍₇₎	92	16.2	3.06
NER-KRVM	4.6	17 ₍₆₎	36	21.6	4.69
KRVM	5.2	26 ₍₇₎	89	15.3	2.94
NER-PRPVM	4.6	18 ₍₄₎	33	20.2	4.39
PRPVM	5.4	24 ₍₆₎	86	17.3	3.20
NER-PRVM	4.5	17 ₍₄₎	31	22.0	4.89
PRVM	5.3	25 ₍₇₎	82	14.3	2.70
POET	7.0	33 ₍₃₇₎	75	16.2	2.31
PR-POET	4.7	26 ₍₆₎	50	16.8	3.57
$p=73$					
NER-TSRVM	4.0	11 ₍₃₎	22	14.9	3.72
TSRVM	141.7	238 ₍₅₀₈₎	4708	-456.5	-3.22
NER-MSRVM	4.0	11 ₍₄₎	23	14.4	3.6
MSRVM	137.5	260 ₍₅₁₄₎	4215	-474.5	-3.45
NER-mMSRVM	4.2	12 ₍₄₎	24	14.3	3.40
mMSRVM	157.8	251 ₍₅₀₇₎	4917	-460.9	-2.92
NER-KRPVM	4.0	10 ₍₃₎	21	14.9	3.72
KRPVM	137.5	236 ₍₄₆₉₎	4352	-413.5	-3.01
NER-KRVM	3.9	11 ₍₅₎	21	14.8	3.79
KRVM	140.6	229 ₍₄₃₁₎	4623	-398.7	-2.83
NER-PRPVM	3.9	10 ₍₂₎	22	15.9	4.08
PRPVM	125.2	224 ₍₄₅₃₎	4145	-416.7	-3.32
NER-PRVM	3.8	9 ₍₂₎	21	16.7	4.39
PRVM	128.1	216 ₍₄₄₈₎	4007	-398.5	-3.11
POET	62.7	188 ₍₂₃₂₎	3116	-235.4	-3.75
PR-POET	4.1	16 ₍₆₎	33	14.5	3.53

Table 1: Empirical results for two sets of stocks with $p = 26$ and $p = 73$ in NYSE respectively. From left to right: Annualized out-of-sample standard deviation, averaged maximum absolute weight, maximum of maximum absolute weight, annualized portfolio return and Sharpe ratio.

where $\mathbf{D}_m^s$ is a diagonal matrix with either 0 or 1 as elements. The j th diagonal element is 1 if the j th asset is already traded at time $t_s^{j_m}$, and 0 otherwise. The matrices $\mathbf{A}(\cdot, \cdot)$ and $\mathbf{\Sigma}(\cdot, \cdot)$ are as defined in Assumption (D1) and (V1) respectively.

(A3) If the drift $\boldsymbol{\mu}_t$ is random, the components of $\mathbf{Z}_{d,s}^j(m+1)$, $\mathbf{Z}_{v,s}^j(m+1) \in \mathcal{F}_{t_s^{j_{m+1}}}^j$ are conditionally independent given $\mathcal{F}_{-j}$, $E(\mathbf{Z}_{d,s}^j(m+1)|\mathcal{F}_{-j}) = \mathbf{0} = E(\mathbf{Z}_{v,s}^j(m+1)|\mathcal{F}_{-j})$, $\text{var}(\mathbf{Z}_{d,s}^j(m+1)|\mathcal{F}_{-j}) = \mathbf{I}_p = \text{var}(\mathbf{Z}_{v,s}^j(m+1)|\mathcal{F}_{-j})$ almost surely. Eighth moments exist for their components as well. If the drift $\boldsymbol{\mu}_t$ is non-random, then $\mathbf{Z}_{d,s}^j(m+1) = (1, 0, \dots, 0)^T$.

(A4) (Only for random drift and volatility). Using notations in Assumption (D2), we assume that for some $c_{d,j,s} \in \mathcal{F}_{-j} \cup \mathcal{F}_s^j$ greater than 0, and for $\ell = 1, \dots, m$,

$$\begin{aligned} & E\left(\mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(\ell+1) | \mathcal{F}_{-j} \cup \mathcal{F}_{t_s^j}^j\right) \\ &= \left(1 - \frac{c_{d,j,s}}{(p-b_s-1)^\alpha}\right) \mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(\ell) + e_{d,s}^{ij}(\ell), \end{aligned}$$

where $0 \leq \alpha \leq 1/2$, and we define $\mathbf{Z}_{d,s}^j(\ell) \mathbf{Z}_{d,s}^j(\ell)^T = \mathbf{I}_p$ and $e_{d,s}^{ij}(\ell) = 0$ for $\ell \leq 0$. The process $\{e_{d,s}^{ij}(\ell)\}$ with $e_{d,s}^{ij}(\ell) \in \mathcal{F}_{t_s^j}^j$ has $E(e_{d,s}^{ij}(\ell) | \mathcal{F}_{-j} \cup \mathcal{F}_{t_s^{j_{\ell-1}}}^j) = 0$ almost surely, and $e_{d,s}^{ij}(\ell) | \mathcal{F}_{-j} \cup \mathcal{F}_{t_s^{j_{\ell-1}}}^j = O_P(\|\mathbf{A}(t_s^{j_{\ell-1}}, t_s^{j_\ell})\|) = O_P(p_f^{1/2} \cdot (p-b_s-1)^{-1} n^{-1} L^{-1})$.

We also have $E\left(\mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{\Sigma}(t_s^{j_m}, t_s^{j_{m+1}})^{1/2} \mathbf{Z}_{v,s}^j(\ell+1) | \mathcal{F}_{-j} \cup \mathcal{F}_{t_s^j}^j \cup \mathcal{F}_{v,s}^\sigma\right) = 0$, since $\{\mathbf{Z}_{v,s}^j(\ell)\}_{s,\ell}$ for $s = 1, \dots, nL$, $\ell = 1, \dots, p-b_s-1$ is a double array of independent random vectors.

(A5) (Only for random drift and volatility). Let $\varphi(x) = e^{x^2} - 1$. We assume that for $\ell = 1, \dots, m$,

$$\begin{aligned} & E\left\{\varphi\left(\frac{|\mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(\ell)|}{|\mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(\ell-1)|}\right) \middle| \mathcal{F}_{-j} \cup \mathcal{F}_{t_s^j}^j\right\} < \infty, \\ & E\left\{\varphi\left(\frac{|e_{d,s}^{ij}(\ell)|}{|\mathbf{p}_{ij}^T \mathbf{D}_m^s \mathbf{A}(t_s^{j_m}, t_s^{j_{m+1}}) \mathbf{Z}_{d,s}^j(\ell-1)|}\right) \middle| \mathcal{F}_{-j} \cup \mathcal{F}_{t_s^j}^j\right\} < \infty. \end{aligned}$$

The assumption for the volatility runs parallel to the above, with the expectations now conditional on $\mathcal{F}_{-j} \cup \mathcal{F}_{t_s^{j_{\ell-1}}}^j \cup \mathcal{F}_{v,s}^\sigma$, $\mathbf{\Sigma}(\cdot, \cdot)^{1/2}$ replaces $\mathbf{A}(\cdot, \cdot)$, $\mathbf{Z}_{v,s}^j(\cdot)$ replaces $\mathbf{Z}_{d,s}^j(\cdot)$ and 0 replaces $e_{d,s}^{ij}(\cdot)$.

Assumptions (A3), (A4) and (A5) are parallel to (D1), (D2) and (D3) respectively. The major difference is that the coefficients $\rho_{d,K,q}^j < 1$ is now replaced by coefficients that are going to 1 as $n, p \rightarrow \infty$, which are controlled by an exponent $\alpha > 0$. A larger α means that the correlations among variables between tick-by-tick trading times are higher. We assume this since the time length between ticks is usually very small. Note that if the drift is non-random, we only need Assumption (A3) that $\mathbf{Z}_{d,s}^j(m+1) = (1, 0, \dots, 0)^T$, which is just a matter of notation rather than an assumption.

Supplementary materials

Please visit <http://stats.lse.ac.uk/lam/Supp-LICOVcomp.pdf> for detailed proofs of Theorems 1, 2, 3 and 4.

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